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Automatic generation of Haiku using Large Language Models: comparing AI creativity with human poetry

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In Japanese, there is a saying, 一期一会 (ichigo ichie), which reminds us that every encounter is unique and happens only once.

I am deeply grateful for all the moments and encounters that have shaped my journey and allowed me to experience life as I have. I am thankful for the people who are still by my side, as well as for those who are no longer part of my everyday life. Each of them has left something within me, and I carry a small part of them wherever I go.

Abstract

This study aims to explore the creative potential of contemporary large language models (LLMs) by examining their ability to generate haiku, a traditional form of Japanese poetry. Using a few-shot prompting strategy, Japanese haiku were generated through an autoregressive text generation pipeline adapted for different open-source models (Gamma, Gemma, LLaMA2 and LLM-JP-Instruct4). To broaden the comparison, additional haiku were produced directly through ChatGPT-5 and Gemini. These AI-generated haiku were then combined with human-written ones and presented in a questionnaire distributed to students at Japanese universities in Tokyo. The survey aimed to assess whether respondents could distinguish between AI-generated and human-written haiku, and to identify the main features and cues on which they relied when making their judgements, using a combination of qualitative and quantitative questions. Recognition accuracy varied across models. ChatGPT-5, Gemini, and Gamma performed at chance level (≈ 0.50), while Gemma, LLM-JP-Instruct4, and LLaMA2 showed moderate detectability (≈ 0.59 – 0.67). However, recognition was strongly item-dependent. Impression ratings predicted perceived humanness but not correct classification, indicating a halo-like attribution bias, and revealing a dissociation between aesthetic evaluation and true authorship detection. As LLMs improve, surface-level creative plausibility may outpace reliable human discrimination, raising questions about how authorship and creativity are cognitively evaluated.

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Chapter 1

Introduction

1.1 AI and Creative Production

Research on computational creativity has a long history, but interest in evaluating AI creativity has intensified significantly following the emergence of Large Language Models (LLMs), particularly in generative systems that operate in natural language [20]. This shift has moved the field from questions about technical feasibility to more nuanced questions about evaluation: What do we mean by creativity in the context of machine-generated text? How should human judges evaluate it? And which aspects of model design most strongly influence perceptions of creativity?

Recent advances in LLMs are grounded in the Transformer architecture introduced by Vaswani et al. (2017) [38], which relies on self-attention mechanisms to model contextual dependencies efficiently at scale. Building on this architecture, models such as **BERT** [10] introduced bidirectional contextual representations trained with masked language modeling objectives, enabling strong performance in language understanding tasks. Subsequently, **GPT** [28] and other decoder-based models adopted an autoregressive training objective, focusing on next-token prediction to generate coherent and contextually consistent text. Larger-scale systems such as **PaLM** [6] and **LLaMA2** [36] further extended this paradigm by scaling model parameters and training data, demonstrating strong generalization capabilities across a wide range of natural language tasks [30].

As generative AI systems gradually become part of everyday life, the question is no longer whether they can produce human-like outputs, but rather whether they can generate creations that humans judge creatively meaningful. Empirical studies comparing human-made and AI-generated poetry have begun to explore this issue [16]. At the same time, research has explored methods to improve generative creativity (e.g., prompt design, decoding strategies, and fine-tuning approaches [20]), but systematic evaluations that link model-level generative behaviour to human aesthetic judgments remain limited.

1.2 Why Poetry? Why Haiku?

Poetry provides a particularly informative domain for studying computational creativity because it combines formal constraints, cultural conventions, and subtle aesthetic judgment. Unlike many other forms of text generation, poetry requires models to produce not only grammatically coherent language, but also evocative imagery and emotional resonance, while operating within stylistic traditions. These characteristics make poetry a setting that is both challenging and revealing for evaluating the creative capabilities of generative models.

Among poetic forms, haiku represents an especially suitable case study. Haiku is a traditional Japanese poetic form typically composed of seventeen morae arranged in a 5–7–5

structure. Despite its brevity, it relies on a number of aesthetic conventions, such as seasonal references and juxtaposing images, which create meaning through suggestion rather than explicit description. The combination of these strict structural constraints with interpretive openness makes haiku an interesting subject for computational analysis. A more detailed discussion of the structure and aesthetic conventions of haiku is provided in Section 2.4.

For computational experiments, haiku offers a useful balance between formal regularity and expressive variation. Its fixed structural constraints provide a controlled generation space in which outputs can be systematically compared across models. Meanwhile, its reliance on imagery and suggestion allows for meaningful differences in tone, interpretation and perceived creativity.

Although prior studies have examined the aesthetic evaluation of AI-generated poetry and compared human- and machine-authored texts [16], and other research has focused on improving generative creativity in large language models [20], relatively little attention has been devoted to a comparative analysis that simultaneously evaluates the linguistic quality of generated haiku across multiple contemporary models and the relationship between perceived creativity and correct authorship attribution. In particular, while previous work has typically focused either on aesthetic judgments or on model-improvement techniques, a systematic cross-model investigation within a formally constrained setting remains limited. It is therefore unclear whether the cues that drive perceived humanness and creativity in generated text are also diagnostic for correct authorship attribution, and how robust these judgments are across different generative systems.

1.3 Research Questions

To address this gap, the present study asks two primary questions:

- **RQ1:** How do contemporary LLMs compare in their ability to generate linguistically and poetically plausible Japanese haiku under the same prompting regimen?
- **RQ2:** How accurately can human readers distinguish AI-generated haiku from human-authored haiku, and does this accuracy differ systematically across models?

1.4 Contributions of the Study

This thesis makes the following contributions:

1. A controlled cross-model evaluation of haiku generation across six contemporary models (ChatGPT-5, Gemini 2.5 Pro, Gemma-2-2b-jpn-it, Japanese-StableLM-Instruct-Gamma-7B, LLaMA2, LLM-JP-3.1-1.8B-Instruct4), contrasted with human-authored haiku drawn from online haiku contests.
2. An empirical study (N = 144 Japanese university students) measuring authorship attribution accuracy and multi-dimensional impression ratings (fluency, coherence, haiku-like wording), allowing analysis of the relationship between perceived creativity and correct identification.
3. A methodological pipeline for haiku generation using few-shot prompting with the Modern Haiku Dataset [27] (37,158 haiku in JSONL format) and a reproducible evaluation protocol that can be extended to other compact poetic forms.
4. New insights into the distinction between aesthetic impression and authorship detection: results show that participants' authorship judgments are often close to chance

for the strongest models, and impression ratings are more closely tied to perceived source than to true source, indicating a dissociation between perceived creativity and correct attribution.

For each of the four seasons (spring, summer, autumn, winter) we generated three regular haiku per season from each model, producing a balanced set of machine-authored items. These were mixed with three human-written haiku per season obtained from public contest sources, and the full set was presented to 144 participants via a questionnaire. Participants judged each haiku as AI- or human-authored and provided ratings on multiple 1–5 scales (e.g., fluency, coherence, haiku-likeness). The generation procedure used few-shot prompting with examples drawn from the Modern Haiku Dataset to reduce variance attributable to prompt design. Statistical analyses examine authorship accuracy, item-level variability, impression score differences, and associations between perceived source and ratings.

1.5 Key Findings

The results suggest that discrimination between human and machine haiku is weak for several of the strongest models: authorship judgments are often close to chance and are highly item-dependent rather than reflecting a stable model-wide signal. Impression ratings (how fluent, coherent, or poetic a haiku appears) track participants' beliefs about the source more than the true source, suggesting that perceived creativity is influenced by inferred authorship. These outcomes point to a conceptual separation between aesthetic impression and correct authorship identification and highlight the importance of item-level analysis in evaluation studies.

1.6 Thesis Overview

The remainder of the thesis is organized as follows. Chapter 2 reviews the literature on computational creativity, large language models, and evaluation methods. Chapter 3 discusses previous work on AI-generated poetry, creativity evaluation, and human versus AI text attribution. Chapter 4 describes the Modern Haiku Dataset, model selection, prompting strategy, and the haiku generation pipeline. Chapter 5 presents the experimental design for the human evaluation, data collection procedures, and analysis plan. Chapter 6 reports the descriptive results of the questionnaire study. Chapter 7 presents the inferential analyses. Chapter 8 examines the qualitative evidence emerging from the open-ended responses. Chapter 9 provides a comparative analysis between human judgments and Grok as an external automated judge. Chapter 10 discusses the broader implications of the findings for computational creativity and authorship evaluation. Finally, Chapter 11 concludes the thesis by summarizing the main findings, theoretical implications, and future research directions.

In sum, this study leverages a formally constrained poetic form to probe the creative capabilities of contemporary LLMs and to investigate how human judgments of creativity and authorship interact in the perception of short, culturally embedded poetic artifacts.

Chapter 2

Background

2.1 Large Language Models and Text Generation

Large language models (LLMs) have recently transformed the field of natural language generation. Most contemporary LLMs are based on the Transformer architecture introduced by Vaswani et al. [38]. This architecture relies on self-attention mechanisms to model relationships between tokens within a sequence. Unlike earlier sequence models based on recurrent architectures, Transformers can process information from different positions in the input in parallel while still capturing long-range dependencies in text.

A key component of the Transformer architecture is the self-attention mechanism, which enables the model to dynamically assign relevance weights to different tokens when generating a sequence representation. Rather than processing tokens sequentially, the model computes attention scores between all token pairs, enabling it to capture contextual relationships throughout the input. This mechanism allows Transformers to efficiently model long-range dependencies that were difficult to capture using earlier recurrent neural networks [38].

Modern LLMs are typically trained using self-supervised learning objectives on large-scale corpora, most commonly next-token prediction. In this autoregressive setting, the model learns to estimate the probability distribution of the next token based on the preceding context. By being exposed to vast amounts of text, the model gradually acquires statistical representations of linguistic patterns, semantic relationships, and discourse structure.

Consequently, LLMs can generate coherent and contextually appropriate text in many different areas, such as dialogue, summarisation, translation and creative writing. Although these systems were not originally designed for artistic applications, their ability to produce fluent language has led to their use in domains such as storytelling, poetry generation, and other forms of creative expression [16, 20].

However, fluency alone does not necessarily equate to creativity, a concept that is deeply human. Text generated by LLMs may be grammatically correct and stylistically consistent yet lack novelty or aesthetic value. Consequently, the ability of LLMs to produce language that is perceived as creative raises important questions about how this quality should be defined, measured, and understood when evaluating outputs produced by artificial systems.

2.2 Computational Creativity

Research in computational creativity has long emphasized that defining creativity is inseparable from defining how it should be evaluated. One widely accepted conceptual foundation is that creative artefacts should exhibit properties such as novelty, surprise, and

value [4]. However, these properties can emerge through different mechanisms and may be interpreted in different ways depending on the evaluative framework used [29].

Boden distinguishes three principal forms of creativity:

- **Combinatorial creativity**, which produces novel combinations of familiar ideas.
- **Exploratory creativity**, which searches within an existing conceptual space.
- **Transformational creativity**, which modifies the conceptual space itself so that previously unattainable ideas become possible.

These distinctions highlight that creative novelty might arise either through recombination or through structural change within a conceptual system [4].

While artificial systems are becoming increasingly capable of modelling generative mechanisms that resemble combinatorial or exploratory creativity [4, 8], evaluating outputs remains a significant challenge in computational creativity research. It is difficult to determine whether a system's output should be considered genuinely creative, as this requires criteria that go beyond simple novelty or statistical variation.

Ritchie proposes that questions about program creativity can be addressed empirically by defining quantifiable relationships between its inputs, processes, and outputs. By defining explicit evaluation criteria, researchers can analyse whether a system's behaviour satisfies certain conditions associated with creative production, rather than relying solely on subjective impressions [29].

Beyond the properties of the artefact itself, several frameworks emphasize the role of perception and context in creativity evaluation. Colton argues that creativity in computational systems should be distinguished from how observers perceive creativity. Even if a system generates artefacts that satisfy the formal criteria for creativity, human evaluators may interpret them differently depending on how the creative process is presented. To capture this multidimensional perspective, the concept of the creative tripod, which describes creativity in terms of three components, is introduced: **skill** (technical competence), **appreciation** (the ability to evaluate artefacts), and **imagination** (the capacity to generate novel ideas) [7].

Similarly, Colton and Wiggins propose a working definition of computational creativity which explicitly incorporates observer evaluation and acknowledges the possibility of human bias in denying creativity to software systems. [8]. This observation highlights the importance of carefully designed experimental frameworks when studying human responses to AI-generated artefacts.

2.3 Authorship Attribution

A persistent challenge when evaluating creative artefacts is the influence of contextual information on human judgement. The intrinsic qualities of an artefact do not determine evaluations alone, as knowledge about its origin may also influence perception. For example, when observers believe that a work was created by a human or by an artificial system, their perception of the same artefact can change significantly.

Studies in visual art provide empirical evidence for this effect. Chiarella et al. demonstrate that attributing abstract paintings to human or AI can influence aesthetic appreciation, with a negative bias emerging particularly when participants are explicitly informed about the role of artificial intelligence in the creative process [5].

Similarly, Bara demonstrates that the way in which AI-generated visual art is presented affects both moral and aesthetic judgements of it. Participants' evaluations vary depending on how the creative process is described, and on whether AI is presented as an autonomous creator or a tool used by humans [3].

These findings suggest that perceived authorship may be an independent factor in aesthetic evaluation. In experimental settings, contextual cues about authorship can influence judgements, even when the artefacts themselves remain identical. This has significant implications for the study of computational creativity, raising questions about how human observers interpret and evaluate AI-generated content when information about the author is ambiguous or absent.

2.4 Haiku: Structure, Aesthetic Conventions, and Relevance for Computational Creativity

In this section, we outline the structure, history, and aesthetic conventions of haiku, drawing primarily on Shirane [32], whose analysis provides a concise overview of the form's historical development and poetic conventions.

Poetry provides a particularly informative domain for studying computational creativity because it combines formal constraints, cultural conventions, and subtle aesthetic judgment. Unlike many other forms of text generation, poetry requires models to produce not only grammatically coherent language, but also to operate within stylistic traditions and structural rules, all the while conveying evocative imagery or emotional resonance. Haiku is an especially suitable case study among poetic forms due to its brevity, recognisable structure, and strong cultural grounding.

Haiku is a traditional Japanese poetic form characterized by extreme concision. In its most widely recognized regular form, a haiku consists of seventeen morae¹ arranged in a 5–7–5 pattern. Despite its structural simplicity, haiku is capable of conveying complex aesthetic effects through minimal language. The apparent paradox of such a short form functioning as poetry has long attracted scholarly attention.

Historically, haiku evolved from the **hokku** (発句), the opening verse of **haikai no renga** (連歌), a collaborative form of linked verse composed by groups of poets. In this context, the opening verse would establish the setting and thematic orientation for the poetic sequence that followed. Over time, particularly during the modern period, the hokku gradually came to be recognised as an autonomous poetic form. The modern term *haiku* (俳句) was popularized in the late nineteenth century by the poet Masaoka Shiki (正岡子規), who emphasized the literary value of the seventeen-mora opening verse as an independent work.

Beyond its metrical structure, classical haiku is deeply rooted in a network of aesthetic conventions that guide both composition and interpretation. One of the most important of these conventions is the presence of a seasonal reference, typically expressed through a **kigo** (季語), or seasonal word. The *kigo* situates the poem within a particular time of year, typically spring, summer, autumn, or winter, and evokes a wider range of cultural associations related to nature, climate, and human activity. In traditional Japanese poetic practice, seasonal words are catalogued in specialized almanacs known as **saijiki** (歳時記), which list established seasonal terms alongside their conventional meanings and literary associations. Through these shared associations, even a single seasonal word can convey complex emotional or cultural implications.

Another defining feature of haiku is the **kireji** (切字), often translated as a “cutting word”. It introduces a pause or conceptual break within the poem, effectively dividing the verse into two juxtaposed parts. This structural division enables the poem to present two images, perceptions or moments that resonate with one another without an explicit explanation being provided. The meaning often emerges from the relationship between

¹Morae are timing units in Japanese phonology and that do not always correspond to syllables in other languages.

these juxtaposed elements, and readers are expected to actively interpret the connection between them. In this sense, the haiku is not a completely closed text; the poem creates an interpretive space that must be filled in by the reader.

The brevity of haiku also shapes its aesthetic logic. As the poems contain only a few words, they cannot provide detailed descriptions or narrative development. Instead, the verse typically presents fragments of a scene or moment, leaving significant interpretive space for the reader to reconstruct the broader situation. Scholars have therefore compared haiku to sketches or snapshots, in which a few carefully chosen details evoke a wider emotional or experiential context. Haiku operate through suggestion, resonance and juxtaposition rather than explicit explanation.

This reliance on cultural associations and reader participation also highlights an important aspect of haiku as a creative system. Although the poem itself is very short, it is connected to a much larger network of shared knowledge and cultural memory. Seasonal words, for example, are rooted in long poetic traditions that link specific natural imagery to emotional or symbolic meanings. Consequently, interpreting a haiku depends not only on the words themselves, but also on the reader's familiarity with these cultural conventions.

However, the global diffusion of haiku has also introduced important transformations. As it spread beyond Japan, the form was adapted by many poets to suit different linguistic and cultural contexts. One significant consequence of this is that the traditional system of seasonal words has often lost its original function. In Japan, these terms are embedded in an extensive cultural and literary tradition, as recorded in seasonal almanacs (*saijiki*), where each word carries well-established poetic associations. Outside Japan, however, these cultural references are typically absent or substantially altered, meaning that the seasonal word does not evoke the same shared meanings for readers. Consequently, many poems written in other languages that adhere to the seventeen-syllable pattern share a similar structure to haiku, yet lack some of the traditional conventions that have historically defined the genre.

A classical example illustrates these characteristics. A well-known haiku by Matsuo Bashō, displayed at the Rakushisha hermitage in Kyoto, reads:



Figure 2.1. A *kuhi* (句碑), or haiku monument, engraved with Bashō's poem. Translation follows.



Figure 2.2. Entrance of Rakushisha (落柿舎), a hermitage in Kyoto historically associated with Matsuo Bashō and his disciple Mukai Kyorai where several haiku monuments are located.

五月雨や
色紙へぎたる
壁の跡

Early summer rain
the marks on the wall
where poetry cards were torn away.

In this poem, the seasonal reference appears in the word *samidare* (五月雨), which denotes the early summer rains. The particle や functions as a cutting word, introducing a pause that separates the atmospheric setting from the visual observation that follows. The poem describes the traces left on a wall after decorative poetry cards (*shikishi* (色紙)) have been removed, evoking absence and impermanence through a small visual detail. As in many classical haiku, the scene itself is minimal yet invites reflection beyond what is explicitly stated.

Despite its brevity, the haiku form thus combines strict formal constraints with considerable interpretive openness. The poet must operate within a limited structural space while selecting imagery capable of evoking broader emotional or cultural meaning, while the reader participates in constructing the poem's interpretation by linking the juxtaposed elements and drawing on shared associations.

These characteristics make haiku particularly suitable for computational experimentation. The fixed structure of seventeen morae provides a clear formal constraint that can be enforced during generation, while the reliance on imagery, cultural context, and reader interpretation introduces aesthetic dimensions that remain difficult for artificial systems to reproduce consistently. Haiku therefore offers a controlled yet expressive setting in which to examine how contemporary language models generate poetry and how their outputs are perceived by human readers.

Chapter 3

Related Work

3.1 AI-generated Poetry

Poetry has long been a popular subject for studying computational creativity due to its combination of linguistic structure and aesthetic judgement. Prior to the emergence of Large Language Models, neural approaches to poetry generation frequently employed recurrent architectures and iterative refinement strategies.

For example, Yan’s iPoet system models the process of generating poems as one that can be refined through multiple iterative steps. This allows earlier drafts to be improved as the poem is created. [41]. Other studies have explored the use of adversarial approaches in creative text generation. For example, Hirota et al. applied Sequence Generative Adversarial Network (SeqGAN) techniques to haiku generation, demonstrating that adversarial objectives can improve stylistic coherence. [15].

More recent research has examined the creation of Japanese haiku using deep neural networks and autoregressive language models that are adapted to the fixed-form constraints and seasonal vocabulary typical of the genre [14, 40].

3.2 Creativity Evaluation in Language Models

Alongside work on generation, researchers have explored methods for evaluating the quality and creativity of AI-generated poetry. Kikuchi et al. propose a neural approach to estimating the quality of haiku using proxy signals such as popularity or community feedback. This represents an early attempt to operationalize aesthetic evaluation computationally [31].

Human evaluation remains central to many studies of computational creativity. For example, Hitsuware et al. (2023) conducted a controlled experiment in which participants evaluated haiku written by humans, generated by AI, or created through human–AI collaboration, and also attempted to identify the authorship of each poem, assessing them along multiple aesthetic dimensions. The results showed that collaborative haiku received the highest beauty ratings, and that participants were generally unable to reliably distinguish between human and AI authorship. [16].

More recently, Tomizawa proposed an approach in which a Large Language Model is used as an evaluator through prompt engineering. In this framework, the LLM acts as a judge whose evaluations can be compared with human assessments or community signals such as contest outcomes [35].

3.3 Human vs AI Text Attribution

Despite progress in generation and evaluation methods, relatively little research has examined the relationship between perceived creativity and authorship attribution. While existing research often investigates aesthetic evaluation or improvements in generative models, it rarely examines how human observers interpret AI-generated texts when the author is unknown.

Studies on aesthetic bias suggest that perceived authorship can have a significant impact on the evaluation of a work[3, 5]. However, comparable analyses in the domain of poetry, particularly within formally constrained genres such as haiku, remain limited.

In particular, few studies compare multiple contemporary generative models while simultaneously examining aesthetic judgments, linguistic evaluation, and authorship attribution accuracy. Controlled experiments in which AI-generated and human-written haiku are mixed, rated along multiple dimensions, and evaluated for authorship attribution allow these factors to be studied together. These designs make it possible to explore the relationship between evaluative cues and perceived creativity, as well as the ability to correctly identify the origin of a text.

Chapter 4

Haiku Generation Pipeline

To investigate the creative capabilities of contemporary language models within the confines of a structured poetic form, we developed a structured haiku generation pipeline. This combines a curated Japanese haiku corpus, controlled prompting strategies and automated filtering procedures in order to generate poems that adhere to the formal characteristics of haiku and can be compared across models.

Haiku were chosen as a controlled experimental setting because they combine strict formal constraints with a high degree of interpretive freedom. The traditional 5-7-5 mora structure and the use of seasonal words (*kigo*) provide explicit constraints that can be verified computationally. However, the poetic effectiveness of a haiku also depends on imagery, juxtaposition and cultural associations, all of which remain difficult for artificial systems to consistently reproduce.

The generation framework therefore focuses on producing structurally plausible haiku while enabling the comparison of different language models under similar prompting conditions.

4.1 Dataset Construction and Filtering

In this study, we used the *Modern Haiku Dataset* [27] available through the Hugging Face repository. This dataset comprises 37,158 haiku written by 4,625 Japanese poets, and it is formatted in JSONL. Each entry includes the poem text alongside metadata fields such as author, source, seasonal word (*kigo*), and the poem’s associated season.

Field	Description
Haiku metadata	
<i>id</i>	ID number of the haiku.
<i>haiku</i>	Text of the haiku.
<i>author</i>	Name of the author of the haiku.
<i>foreword</i>	Additional introductory text. Nullable.
<i>source</i>	Name of the source document of the haiku. Nullable.
<i>comment</i>	Comment about the haiku written by a reviewer. Nullable.
<i>reviewer</i>	Name of the reviewer who made the comment. Nullable.
<i>note</i>	Additional note about the haiku. Nullable.
<i>season</i>	Season associated with the haiku. If the haiku has no seasonal word, this field is set to none.
Kigo metadata	
<i>id</i>	ID number of the seasonal word.
<i>word</i>	The seasonal word itself.
<i>kana</i>	Pronunciation of the word.
<i>old_kana</i>	Pronunciation in old hiragana orthography. Nullable.
<i>season</i>	Season associated with the seasonal word.
<i>subtitle</i>	Alternative name of the seasonal word.

Table 4.1. Structure of the haiku dataset and associated metadata fields.

All of the poems in the dataset are written in Japanese. We intentionally conduct the experiments in the original language rather than using translated haiku. As discussed in Section 2.4, key properties of haiku, such as the mora-based 5–7–5 structure and the use of seasonal words (*kigo*), are deeply embedded in the linguistic and cultural conventions of Japanese poetry [32]. Translations into other languages typically approximate these features using syllables rather than morae, often losing the cultural associations attached to seasonal vocabulary in the process. Working with the original Japanese text therefore enables us to preserve the structural and cultural characteristics of the haiku form.

Data Filtering

An initial filtering stage was performed before using the dataset for generation experiments. Records with excessively sparse or irrelevant metadata fields were removed. Fields such as *comment*, *reviewer*, and *foreword* were frequently incomplete or unrelated to the poetic content itself, so they were removed. Removing these entries allowed us to retain a corpus focused primarily on the poetic text and essential descriptive metadata.

Mora Structure Labeling

To evaluate the structural properties of the poems, a mora-counting function was used to process each haiku. Based on the number of morae detected in each line, poems were labeled as either *regular* or *irregular*. Regular haiku correspond to the canonical 5–7–5 mora structure, while irregular poems deviate from this traditional pattern. This distinction allows us to analyze the structural conformity of generated poems and to measure whether models reproduce the traditional haiku metric.

Kigo Dictionary Separation

The dataset includes a built-in dictionary of seasonal words (*kigo*). During pre-processing, the kigo dictionary was separated from the main corpus, creating two complementary datasets:

one containing the haiku texts, and the other containing the seasonal vocabulary.

Separating these components allows for more precise control over seasonal references during the generation process, as well as making it easier to identify seasonal words during tokenization.

Orthographic Considerations

During the preliminary inspection of the corpus, we identified the presence of archaic kana characters such as *ㇰ* (*we*) and *ㇱ* (*wi*). These characters belong to pre-war Japanese orthography and were officially removed from the standard spelling system during the 1946 orthographic reforms.

Instead of normalizing these characters by replacing them with their modern equivalents, we chose to keep them in order to preserve the historical authenticity of the poems. Their presence suggests that some entries originate from earlier literary editions that adhere to traditional spelling conventions. These forms are relatively rare, accounting for less than 0.03% of all characters in the corpus, but they reflect the historical depth of the dataset.

Tokenization

The haiku texts were tokenized using MeCab [18] with the UniDic dictionary [9] through the Fugashi Python interface [19]. This configuration is commonly employed in Japanese language processing and offers a level of morphological segmentation that is well-suited for short poetic texts such as haiku.

Each poem was broken down into its basic linguistic units. Then, tokens were compared against the kigo dictionary to identify any seasonal words present in the haiku. When a seasonal word was detected, it was assigned to the relevant season. This process generated a tokenised representation of the poem and an explicit seasonal label.

Finally, during generation experiments, tokenization and decoding were performed using the tokenizer associated with the selected language model to ensure compatibility with the model’s internal vocabulary.

4.2 Model Selection

This study aims to evaluate how contemporary large language models generate poetry within strict structural constraints. To this end, we will focus on transformer-based autoregressive language models that can produce coherent text sequences.

Experiments were conducted using both open-source and closed-source models to enable a comparison between publicly reproducible systems and state-of-the-art commercial models.

Selected Models

The study includes six language models representing different architectures and parameter scales:

- **Gemma 2 JPN** [13]
- **Japanese StableLM Gamma** [34]
- **ELYZA Japanese LLaMA2** [37]
- **LLM-JP-Instruct4** [1]

- ChatGPT-5 [26]
- Gemini 2.5 Pro [12]

Model Architecture

Most contemporary Large Language Models are based on the Transformer architecture introduced by Vaswani et al. (2017) [38]. Transformers use a self-attention mechanism, which enables the model to identify long-range dependencies between tokens without the need for recurrent or convolutional structures. Rather than processing text sequentially, the Transformer architecture computes contextual representations of tokens via stacked attention layers and feed-forward networks.

Transformer-based models can generally be divided into three main architectural configurations: **encoder-only models**, **encoder–decoder models**, and **decoder-only models**.

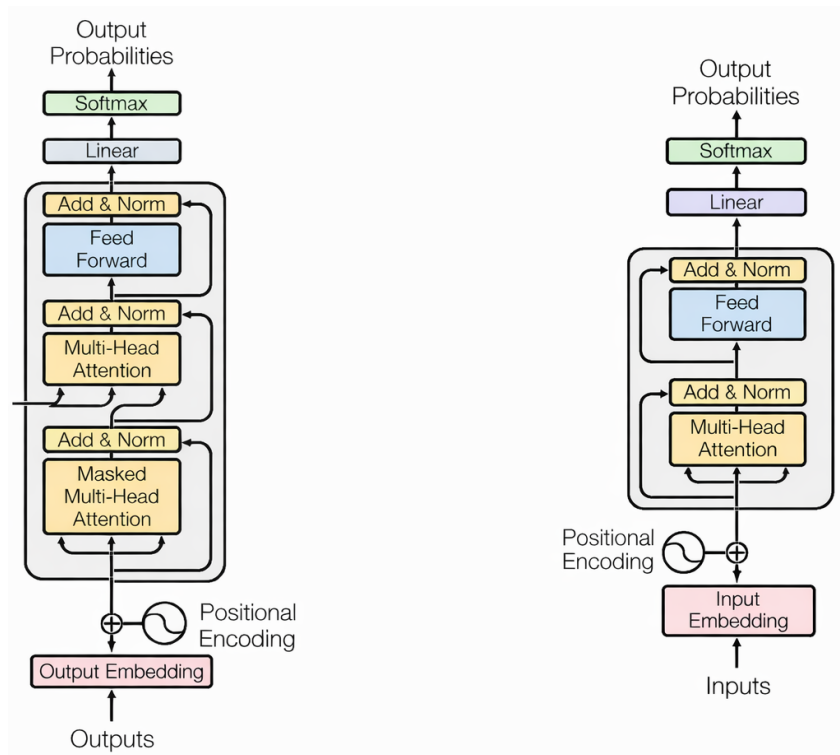


Figure 4.1. Encoder-only architecture on the right and decoder-only architecture on the left. Reproduced with AI from an online [image source](#).

Encoder-only architectures, such as BERT [10], are primarily designed for representation learning tasks such as classification, sentence similarity, or information retrieval. In these models, the encoder processes the entire input sequence simultaneously, producing contextual embeddings for each token.

Decoder-only architectures, which are commonly used in Large Language Models, generate text autoregressively by predicting each token based on the previously generated tokens [28]. During generation, the model produces the output sequentially, conditioning each new token on the context provided by the tokens generated earlier in the sequence. This configuration is particularly well-suited to open-ended text generation tasks and forms the basis of many modern language model families, including GPT- and LLaMA2-style systems [28, 37].

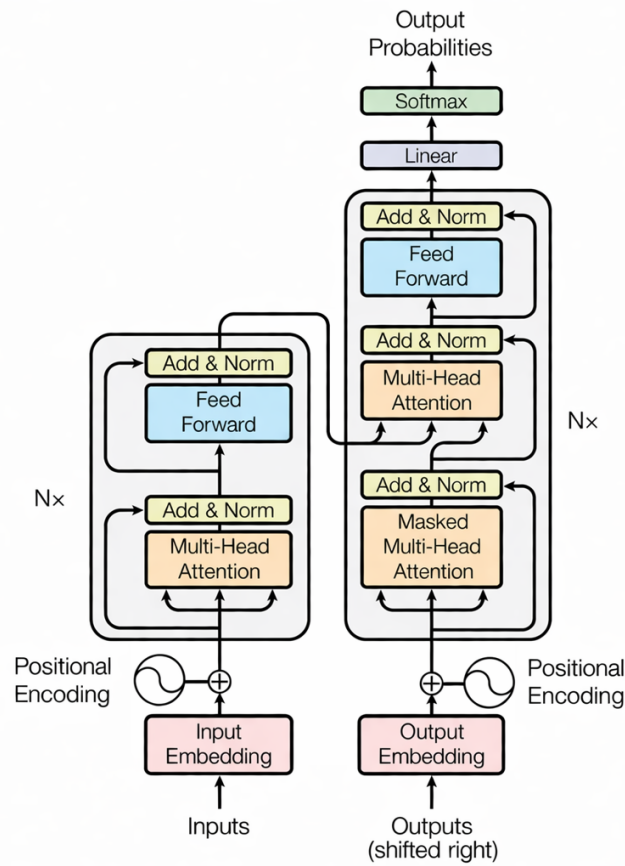


Figure 4.2. The Transformer architecture. Source: [38].

Encoder–decoder architectures, such as the original Transformer proposed for machine translation [38], consist of two components: an encoder that processes the input sequence and a decoder that generates the output sequence. These architectures are commonly employed in sequence-to-sequence tasks, such as translation and summarization.

All the open-source models employed in this study adhere to a decoder-only autoregressive architecture. Specifically, these models generate text one token at a time using causal self-attention, ensuring that each token can only refer to tokens that appear earlier in the sequence.

Closed-source systems such as ChatGPT-5 and Gemini 2.5 Pro are also believed to rely on large-scale transformer architectures, although their internal configurations are not publicly disclosed. For this reason, these systems are treated as black-box generative models and evaluated solely through their output behaviour.

Model Size and Characteristics

Model	Parameters	Source	Japanese Support	Developer	Architecture
Gemma 2 JPN	2B	Open	Native	Google	Transformer
StableLM Gamma	7B	Open	Native	Stability AI	Transformer
LLaMA2	7B	Open	Native	ELYZA	Transformer
LLM-JP-Instruct4	1.8B	Open	Native	NII Japan	Transformer
ChatGPT-5	undisclosed	Closed	Native	OpenAI	undisclosed
Gemini 2.5 Pro	undisclosed	Closed	Native	Google	undisclosed

Table 4.2. Language models used in the haiku generation experiments

During generation, each model was used with its native tokenizer in order to maintain compatibility with its internal vocabulary and decoding procedure.

4.3 Few-Shot Prompting Strategy

Initial experiments explored a fine-tuning approach before adopting prompting-based generation. The Gemma-2b-jpn model was fine-tuned using Low-Rank Adaptation (LoRA) [17], with the hyperparameters optimized through a series of Optuna trials conducted on progressively larger subsets of the haiku corpus. The experiments explored variations in learning rate, scheduler type, batch size, number of epochs, and LoRA configuration.

During the tuning process, the dataset was refined to improve structural consistency. From early trials onwards, the corpus was filtered to retain only haiku that adhered to the standard 5–7–5 mora structure. Even when their labels were valid, irregular poems were removed, as they introduced structural noise that made it harder for the model to learn the canonical form of haiku. Following this filtering step, the generated poems demonstrated greater adherence to formal constraints and a more consistent poetic structure.

Despite these improvements, the study ultimately shifted towards prompt-based generation strategies. This allowed the model to produce new outputs without altering its parameters, thereby facilitating the testing of multiple models and decoding strategies under comparable conditions.

Few-shot prompting was adopted as the main generation strategy in order to guide the models toward producing haiku that resemble the structure and style of the reference corpus without modifying the model parameters. With this method, a few example haiku are included in the prompt, enabling the model to infer the desired format, stylistic conventions and structural constraints from the provided examples.

This strategy was chosen because it is lightweight, interpretable, and well suited to comparative experiments across multiple models. It also makes it possible to control important generation cues, such as seasonal vocabulary, poetic structure, and stylistic tone through prompt design rather than through additional training.

Prompt Template

All prompts were written in Japanese to match the language of the corpus and preserve Japanese haiku’s linguistic and cultural conventions. The prompt was designed to give the model explicit information about the seasonal word, (*kigo*), the season, and the expected poetic structure.

Each example block followed the same template:

Listing 4.1. Example prompt construction function for Gemma

```
def build_example_block_gemma(ex):
    return (
        "以下の情報をもとに、美しい日本語の俳句を三行で一つ作って  

        ください。季語と季節を含めてください。\\n\\n"
        f"季語: {ex['word']}\\n"
        f"季節: {ex['season']}\\n"
        f"構造: {ex['haiku_structure']}\\n"
        f"俳句:\\n{ex['ref_haiku']}\\n"
        "<end_of_turn>\\n"
    )
```

Note. Literal translation of the Japanese prompt:

“Based on the following information, please create one beautiful Japanese haiku in three lines. Please include a seasonal word and the season.

Seasonal word: {ex['word']}

Season: {ex['season']}

Structure: {ex['haiku_structure']}

Haiku: {ex['ref_haiku']}

This prompt structure uses explicit instructions, structured input fields and example conditioning to guide the model towards generating haiku that are both formally correct and contextually appropriate.

The target generation block used the same structure but omitted the reference poem, requiring the model to produce a new haiku. Additional structural hints indicating the expected 5-7-5 segmentation were included in the target block when available. These hints acted as soft constraints, encouraging the model to preserve the canonical mora pattern while still allowing poetic variation.

Number of Examples

Few-shot prompting was implemented by including a small number of example haiku before the target request. In the final configuration, six examples were used ($k = 6$). This value was selected empirically following preliminary experiments, reflecting a balance between providing sufficient stylistic guidance and avoiding overly long prompts or excessive copying of examples.

Each example block contained: the **seasonal word** (*kigo*), the **associated season**, the **haiku structure label**, and a **reference haiku** drawn from the dataset.

These examples served as demonstrations of the desired poetic format, encouraging the model to produce new poems with similar structural and stylistic patterns.

Retrieval Strategy	Example Selection	Purpose
Same season + kigo	Exact match	Baseline conditioning
Same season + kigo (filtered)	Excluding target poem	Prevent prompt leakage
Mixed retrieval	Season + kigo mixture	Improve diversity
Semantic k-NN retrieval	Embedding similarity	Context relevance
Season-first retrieval	Prefer 5-7-5 poems	Final configuration

Table 4.3. Few-shot example retrieval strategies explored during development.

Prompt Format

Each prompt consisted of a sequence of example blocks followed by a target generation block. The general structure was:

1. Example block 1
2. Example block 2
3. ...
4. Example block k
5. Target generation block

This format allowed the model to infer the expected output style from the examples, adapting the generation process to the new seasonal term specified in the target request.

Prompt optimisation

Prompt design was refined iteratively through three stages: **manual prompt engineering**, **LLM-assisted prompt exploration**, and **automatic template selection**. A detailed overview of the individual experimental configurations (Prova 1–Prova 18) is provided in Appendix A.

Prompt Strategy	Few-shot k	Retrieval Method	Decoding
Basic template	4	None	Sampling
Optimized prompt	4	None	Beam search
Optimized + semantic retrieval	8	Semantic k-NN	Sampling
Gemma-style structured prompt	5	None	Beam search
Gemma + structural constraints	6	None	Beam search
PromptOptimizer template selection	6	None	Beam search

Table 4.4. Summary of prompt engineering strategies explored during development.

The first stage consisted of manual prompt engineering (Prova 1–Prova 7). Different prompt styles were tested, ranging from minimal instructions (e.g., simply requesting the generation of a three-line haiku) to more structured prompts specifying poetic form, seasonal constraints, and example conditioning. These experiments also explored different decoding strategies, including sampling and beam search. The most effective configuration combined a structured prompt in Japanese with a small set of few-shot examples and explicit instructions regarding the inclusion of a seasonal word (*kigo*) and the three-line format of the poem.

In the second stage (Prova 8–Prova 17), additional prompt variants were explored with the assistance of external large language models. Conversational systems such as ChatGPT [26] and Claude [2] were used to suggest alternative formulations of the task instructions and prompt structure. These suggestions served as starting points for further manual refinement and were subsequently evaluated within the generation pipeline together with variations in decoding parameters and retrieval settings. Although several of these prompts produced outputs comparable to the best manually designed configuration, none consistently improved the generation quality or structural accuracy.

Finally, a small-scale automatic prompt optimisation procedure was tested (Prova 18). A pool of candidate prompt templates was defined and evaluated using an internal scoring function combining BLEU and perplexity. For each template, the model generated candidate haiku which were scored against reference examples, and the template with the highest average score was selected. Although this approach provided a systematic method for comparing prompt structures, the automatically selected templates performed slightly worse than the best manually engineered prompt and were therefore not adopted as the final generation configuration.

Prompt Wording Effects

During preliminary one-shot generation experiments, a minor variation in the wording of the prompt produced a noticeable effect on the quality of the output. The initial prompt used the formulation 「あなたは日本語の俳人です」 (“you are a Japanese-language haiku poet”), which frames the instruction in terms of language rather than nationality. In standard Japanese usage, a more idiomatic formulation would typically be 「あなたは日本人の俳人です」 (“you are a Japanese haiku poet”).

Unexpectedly, the language-framed version consistently produced better results. When applied to StableLM Gamma-7b and Gemma-2b-jpn under identical decoding settings, this prompt produced haiku with richer imagery, stronger seasonal fit, and a more natural poetic tone. In quantitative terms, average perplexity decreased from around 52 to 34 for StableLM Gamma-7b, and from approximately 6,975 to around 530 for Gemma-2b-jpn.

When the wording was changed to the nationality-framed version, generation quality declined. One possible explanation is that instruction-tuned models are more strongly influenced by patterns such as ‘write X in Japanese’ than by nationality-based cues [39]. Native-speaker feedback indicated that, while somewhat unusual, the language-framed expression is still acceptable. Due to the consistent empirical improvement, the language-based formulation was retained for the one-shot comparisons.

Implementation and Execution

The prompting pipeline was implemented in Python and developed using Visual Studio Code. As the experiments required multiple generation runs across models and parameter settings, the few-shot prompting experiments were executed using the CINECA high-performance computing infrastructure [25].

Jobs were submitted through SLURM batch scripts, enabling automated and reproducible execution of the generation pipeline across different models and prompting configurations.

4.4 Decoding Strategy

Text generation was performed using several decoding strategies that are commonly applied to autoregressive language models. As haiku generation requires a balance of structural control and poetic variation, several decoding approaches were tested before the final configuration was selected.

During development, a large number of prompt and decoding configurations were tested. A summary of these intermediate trials is provided in Appendix A.

Strategy	Generation Mode	Decoding Method
Beam search baseline	Whole haiku	Beam search
Line-by-line beam search	3-step	Beam search
Sampling decoding	3-step	Temperature sampling
Sampling + structural filtering	3-step	Top-p sampling
Sampling + reranking	3-step	Sampling + BLEU/PPL scoring

Notes. Beam search baseline uses deterministic generation.

Line-by-line beam search enforces the 5–7–5 syllable structure.

Sampling decoding increases diversity.

Structural filtering reduces prompt copying.

Sampling + reranking corresponds to the final configuration.

Beam Search Experiments

Early experiments relied on beam search in order to improve structural stability. In particular, diverse beam search was employed to promote variability among candidate generations while preserving coherence. In the one-shot setting, the optimal beam search configuration employed:

```
k = 6
max_new_tokens = 20
num_beams = 6
num_return_sequences = 5
num_beam_groups = 3
diversity_penalty = 0.7
```

Although beam search generally improved formal regularity, it also tended to favor locally high-probability continuations. In practice, this often resulted in the copying of lexical material and n-grams from the few-shot examples, thereby reducing novelty and weakening the creative value of the generated haiku.

Iterative Line-by-Line Generation

To enforce stronger structural control, an **iterative line-by-line strategy** was also tested. In this approach, the haiku was generated one line at a time: first a 5-mora line, then a 7-mora continuation conditioned on the first line, and finally a 5-mora ending conditioned on the previous two lines.

This approach improved control over the 5–7–5 structure and made it easier to ensure the inclusion of a seasonal word. However, when combined with beam search, it often generated outputs that replicated content from the prompt examples. Furthermore, while structural adherence remained high, coherence between lines often deteriorated; the second and third lines sometimes introduced imagery that did not connect well with the opening line, resulting in the poem feeling assembled rather than unified.

For this reason, the initial line-by-line beam-search variant was not retained as the final setting.

Sampling-Based Decoding

To reduce prompt copying and increase novelty, stochastic sampling was introduced as an alternative to beam search. This sampling-based decoding approach enables the model to select from multiple plausible continuations, rather than following the highest-probability path deterministically. This makes it more suitable for creative tasks, such as generating poetry.

The final sampling-based configuration used in the line-by-line setting was:

```
few_shot_k = 6
max_new_tokens = 22
num_targets = 24
do_sample = True
temperature = 0.7
top_p = 0.90
top_k = 50
repetition_penalty = 1.15
```

This configuration maintained strong structural control while reducing the tendency to reproduce the original prompts. Compared with beam search, sampling produced more varied poetic outputs and improved novelty, although this came at the cost of longer runtimes and occasional reductions in inter-line coherence.

Filtering and Structural Constraints

After the generation, candidate haiku were filtered according to several formal constraints. First, the outputs were cleaned to remove extraneous symbols, markup and non-poetic fragments. Then, candidate lines were evaluated using a function that counted morae in order to verify adherence to the target 5–7–5 structure.

Additional filters were applied to detect the presence of the required seasonal word (*kigo*) and to reject outputs containing prompt-like artefacts, metalinguistic elements, or obvious structural violations. Anti-copy mechanisms were also introduced to minimise overlap with the few-shot examples, such as n-gram blocking and similarity checks against reference poems.

Scoring and Reranking

When multiple candidate generations were available, they were re-ranked using a combined score based on BLEU similarity and perplexity. In particular, the reranking criterion followed the form:

$$\text{score} = \text{BLEU} - 0.2 \cdot \log(\text{PPL})$$

The aim of this scoring function was to strike a balance between formal similarity to the reference haiku and the fluency and plausibility captured by the model’s perplexity estimate. Although BLEU is limited in creative tasks such as poetry writing, combining it with structural checks and qualitative inspection provided a useful approximate measure.

Overall, the final decoding strategy involved a combination of stochastic sampling, explicit structural filtering and reranking. This enabled the generation of haiku that were formally constrained yet sufficiently diverse for subsequent human evaluation.

4.5 Closed-Model Baselines

To complement the experiments conducted with open-source models, two closed-source large language models were used to produce additional generations: **ChatGPT-5** and **Gemini 2.5 Pro**.

These models were accessed via their public interfaces rather than a programmable API. Consequently, their outputs were collected manually and primarily used for qualitative comparison rather than systematic benchmarking.

For ChatGPT-5, the model was given a Japanese prompt to generate twelve haiku in the traditional 5-7-5 mora structure. The prompt prohibited the use of numbers, Latin characters, symbols, or explanatory annotations, but permitted the natural occurrence of cutting words (*kireji*). The model was also instructed to incorporate at least one seasonal word (*kigo*) into each haiku.

As the interaction took place via a conversational interface, several iterations of the prompt were required to ensure the generated haiku were returned in the desired format of three lines each, with no additional commentary or formatting errors.

Gemini 2.5 Pro was asked to generate haiku using a similar prompt with identical structural constraints. These trials were not intended to provide a rigorous benchmark, but rather to offer an external point of reference for the quality that can be achieved by large open-source models compared to the smaller open-source models used in the main experiments.

The generated haiku were manually collected and translated into English for analysis.

あなたは日本語の俳人です。数字・英字・記号・注釈は禁止。切れ字は自然に。
新しい5-7-5モーラの12俳句を書いてください。
季語も入れてください。

Figure 4.3. Japanese prompt given to ChatGPT-5 to generate haiku. *Translation: "You are a Japanese haiku poet. Numbers, letters, symbols, and annotations are prohibited. Use cutting words naturally. Please write 12 new haiku in the 5-7-5 syllable structure. Include a seasonal word."*

4.6 Selection of Human Haiku

In order to investigate how contemporary AI-generated haiku are perceived in relation to those written by humans, a selection of human-written haiku were chosen to be included in the evaluation questionnaire.

Human-written haiku were collected from curated online haiku contests hosted by established organizations: *Haiku Nippon* [21–24], *Sophiakai* [33], and the *EJC Foundation* [11]. These sources were chosen because they publish haiku selected by contest judges from open submissions and therefore provide examples of authentic modern haiku written in Japanese.

The decision to rely on contemporary prize-winning haiku rather than canonical poets such as Bashō or Issa was deliberate. While the sample of participants was not limited to literature majors, they were all students enrolled at Japanese universities, for whom canonical haiku may still be culturally familiar. Using highly recognisable classical authors would therefore have increased the risk of recognition bias, as some poems might have been identified through prior familiarity rather than being evaluated solely on their linguistic or poetic qualities.

From these collections, poems were selected according to explicit criteria. Only haiku that adhered to the traditional 5-7-5 *mora structure* and contained a distinct seasonal reference (*kigo*) were retained. To reduce researcher selection bias and maintain consistency with the AI-generated outputs, the first three traditional haiku available for each season were chosen. This yielded a balanced set of twelve human-written haiku, with three for each season.

The selected human poems serve two complementary purposes in the evaluation framework. Firstly, they provide a stylistic baseline that represents contemporary haiku writing. Secondly, they support the authorship classification task, in which participants attempt to distinguish between poems generated by humans and poems generated by machines. By incorporating high-quality contemporary haiku from recognised competition sources, the evaluation design ensures that the comparison reflects realistic poetic standards, while reducing the impact of familiarity with canonical literary authors.

These human-written haiku were later combined with AI-generated poems in the evaluation questionnaire described in the next Chapter.

Chapter 5

Experimental Design

5.1 Questionnaire Structure

This study used a questionnaire-based human evaluation to assess the perceived creativity and authorship of the generated haiku. Human judgement plays a central role in computational creativity research because the aesthetic quality of artistic artefacts depends on human interpretation rather than purely computational metrics.

The design of the questionnaire was informed by prior studies on computational poetry evaluation and haiku generation. In particular, the evaluation protocol draws inspiration from three works: the human evaluation framework proposed by Yan et al. [41], the human study used by Hirota et al. [15], and the evaluation methodology described by Tomizawa et al. [35]. These studies show how structured human evaluation can be used to judge the linguistic quality and perceived creativity of machine-generated poetry.

Yan et al. [41] evaluated automatically generated Chinese poetry using a set of binary criteria judged by human evaluators. Participants rated poems based on their satisfaction of certain poetic properties, such as grammatical correctness and meaningfulness. The final score was obtained by summing the individual judgements. Although the present study does not use a binary scoring system, the decision to rely on questionnaire-based evaluation rather than purely automatic metrics was informed by the use of human judges to assess poetic quality.

Hirota et al. [15] conducted a small-scale study involving Japanese university students evaluating a mixed set of human-written and machine-generated haiku. Participants rated each poem on several aesthetic dimensions, such as haiku-like wording, poetic imagery, understandability and overall favourability, using a Likert scale. After completing the ratings, the participants were asked to judge whether each poem had been written by a human or generated by a machine. This combination of aesthetic evaluation and authorship attribution had a significant impact on the design of the questionnaire used in this study.

Tomizawa et al. [35] proposed a protocol for evaluating haiku poems, in which they were assessed according to several qualitative criteria, including thematic relevance, grammatical correctness, meaningfulness, and coherence between phrases. While their study focused on expert haiku poets rather than student participants, the use of multiple complementary evaluation criteria informed the decision to include several dimensions of impression ratings in the present questionnaire.

Drawing on these previous approaches, the questionnaire was designed to capture both aesthetic judgments and authorship attribution. The survey was implemented as an online questionnaire and made available in Japanese and English. The English version was included primarily to accommodate exchange students or participants with lower Japanese proficiency. At the beginning of the questionnaire, the respondents were asked to select

their preferred language.

Each questionnaire consisted of a sequence of haiku evaluation items. Participants were asked to complete the same set of evaluation questions for each haiku. The evaluation block for each poem included:

- impression ratings across several aesthetic dimensions using a five-point Likert scale;
- an authorship classification task indicating whether the poem was written by a human or generated by AI;
- a confidence rating indicating how certain participants were about their authorship judgment;
- a question asking whether the poem appeared related to the intended seasonal theme;
- an optional open-ended question asking about the general imagery, wordfeel, or reason for the respondent choices.

In addition to the repeated item-level questions, the questionnaire concluded with a small set of optional open-ended questions. Participants were asked to provide overall feedback and to describe the cues that had helped them to distinguish between AI-generated and human-written haiku. They were also asked to indicate which items they had found particularly difficult to evaluate and explain why. In total, each participant evaluated twelve haiku.

To prevent participants from inferring patterns about the origin of the poems, human-written and AI-generated haiku were randomly mixed within each questionnaire. Participants were informed that the survey contained both types of poems, but were not told which poems belonged to which category, nor how many were written by humans and how many were generated by AI. This design ensured that evaluations were blind, based solely on the perceived qualities of the poems.

The original questionnaire contained twenty-four haiku, comprising twelve poems written by humans and twelve poems generated by AI, balanced across the four seasons. However, the pilot study indicated that evaluating all of the poems would take too much time and risk participant fatigue. For this reason, the questionnaire was divided into two versions. One version contained haiku related to spring and summer, while the other included poems associated with autumn and winter. Each participant completed only one version of the questionnaire and was not informed about the existence of the other version. Consequently, each respondent evaluated twelve poems in total, with the questionnaire taking approximately ten to fifteen minutes to complete.

Prior to full distribution, a pilot study was conducted to evaluate the clarity of the questions and effectiveness of the response collection process. Taking place in the Language Processing Laboratory at the Shibaura Institute of Technology in Tokyo at the end of October 2025, the pilot study involved four to five participants and featured both human-written and AI-generated haiku. The human-written haiku were sourced from contests, while the AI-generated haiku were produced during generation experiments.

Several improvements were made to the questionnaire design following feedback from the pilot study. In particular, definitions were added to clarify the meaning of the impression rating categories, since some participants had initially interpreted the evaluation dimensions differently. The pilot study also revealed that participants relied on various interpretive cues when attempting to distinguish between human and AI authorship. Qualitative responses to the open-ended question about authorship clues indicated that readers often considered perceived imagery richness, emotional resonance, naturalness of phrasing, and rhythmic flow when making their judgments.

Although the pilot study was primarily intended to test the structure of the questionnaire, the qualitative responses provided valuable insights into the criteria readers use to evaluate

AI-generated poetry. These observations complement the results of the quantitative evaluation and help to contextualize the patterns observed in the main experiment.

The questionnaire was designed as an online survey and distributed via university courses and academic networks. A public demonstration version of the questionnaire is available online.¹

5.2 Participant Sample

The evaluation involved students from several Japanese universities. Initially, the questionnaire was intended for students in the Literature and Psychology departments at Atomi University. However, as the response rate from this group was lower than expected, the questionnaire was distributed to additional institutions, including Shibaura Institute of Technology, to obtain a sufficient number of participants.

As a result, the final pool of participants included a total of 144 students from both the humanities and STEM disciplines. Although students from literature programmes are likely to be more familiar with Japanese poetry, participants from scientific disciplines tend to have less formal experience of haiku composition. This diversity of academic backgrounds enabled the study to capture a broader range of perspectives when evaluating poetic texts.

In addition to native Japanese students, a small number of exchange students also participated in the survey. Their Japanese proficiency varied, ranging from beginner levels to approximately JLPT N2. Data collection was completed on December 3, 2025.

At the beginning of the questionnaire, participants were asked to provide some basic background information, such as their academic field, their Japanese language proficiency level, and their previous experience with haiku, for example whether they had read or composed any before. These variables were later used to examine whether participant characteristics influenced their ability to recognize or evaluate authorship.

5.3 Impression Rating Dimensions

Participants evaluated each haiku based on a set of impression ratings designed to capture various aspects of poetic quality.

Each dimension was rated using a five-point Likert scale ranging from 1 (very low) to 5 (very high). The seven evaluation dimensions included:

- **Fluency** = grammatical and syntactic naturalness of the poem.
- **Haiku-like wording** = whether the phrasing resembles traditional haiku style.
- **Poeticness** = the degree to which the poem evokes imagery or emotional resonance.
- **Line coherence** = the extent to which the three lines form a coherent poetic unit.
- **Understandability** = whether the meaning of the poem is clear and interpretable.
- **Favourability** = the participant's overall appreciation of the poem.
- **Unexpectedness** = the degree of surprise or novelty conveyed by the poem.

These dimensions were chosen to capture the linguistic and aesthetic aspects of haiku evaluation. This allowed the analysis to explore the relationship between perceived poetic quality and authorship attribution.

¹<https://forms.gle/L4LFLaaxiagYg9927>

5.4 Authorship Classification Task

In addition to providing impression ratings, participants performed an authorship attribution task similar to a Turing-style evaluation. For each haiku, they had to indicate whether they thought it was written by a human or generated by artificial intelligence.

Participants also rated their confidence in this judgment on a five-point scale, ranging from low confidence to high confidence. This additional measure enabled the study to examine not only whether participants had correctly identified the source of a poem, but also how certain they felt about their decision.

Finally, participants were asked whether they thought the haiku was related to the intended theme or seasonal context. This question was used to assess whether the poems successfully incorporated thematic elements, such as seasonal references.

5.5 Seasonal Questionnaire Split (AW / SS)

The initial questionnaire design included 24 haiku: twelve human-written poems and twelve AI-generated poems. The set was balanced across the four seasons, with three poems from each category for each season.

However, the results of the pilot study indicated that completing the full set took too long. In order to reduce participant fatigue and improve completion rates, the questionnaire was therefore divided into two versions.

One version contained haiku related to spring and summer (SS), while the other contained poems related to autumn and winter (AW). Each version therefore included 12 poems in total. Participants completed only one version of the questionnaire and were not informed about the existence of the other version.

On average, completing one questionnaire required approximately 10–15 minutes.

5.6 Data Cleaning and Preprocessing

After the questionnaire responses had been collected, the dataset underwent pre-processing before statistical analysis. Responses that were incomplete or contained missing answers were removed.

In order to facilitate quantitative analysis, Likert-scale responses were converted into numerical values ranging from 1 to 5. Responses regarding authorship attribution were encoded as binary variables representing either human-written or AI-generated content.

Participant metadata, including academic background and language proficiency, were standardized to enable group-level comparisons. These preprocessing steps guaranteed that the responses collected could be analyzed reliably in the subsequent evaluation stage.

Chapter 6

Descriptive Results

This chapter presents the descriptive statistics obtained from the human evaluation experiment. The analysis focuses on impression ratings, authorship recognition accuracy and item-level response patterns. The aim is to summarize the observed data patterns without applying inferential statistical modeling, which is covered in the next chapter.

As the questionnaire was divided into two seasonal versions (spring–summer and autumn–winter), the responses were initially analysed separately for each seasonal group and model configuration. The questionnaires were distributed across six AI models: Gemma, StableLM Gamma-7B, LLaMA2, LLM-JP-Instruct4, ChatGPT-5, and Gemini. For the descriptive analysis, the questionnaires were first grouped into pairs according to the AI model that had been used to generate half of the haiku in each version of the questionnaire.

Prior to the analysis, the datasets were cleaned to remove incomplete or clearly invalid responses. This included missing answers and cases where metadata fields were incorrectly filled in, such as respondents entering their student identification number in the university name field. After cleaning, a preliminary examination of respondent background information was conducted, including university affiliation, language proficiency, academic department and prior experience of reading or writing haiku.

Background analysis showed that the majority of respondents were students from Japanese universities and native Japanese speakers. Most participants also reported having had previous experience of haiku, either through reading or writing them. This suggests that the participants were familiar with the haiku genre, both linguistically and culturally.

The descriptive analysis proceeds in several stages. First, the distribution of impression ratings is examined. Next, the accuracy of overall authorship recognition is reported, followed by model-specific confusion matrices. Finally, the analysis looks at haiku-level accuracy and patterns of item dependence across the dataset.

6.1 Impression Ratings

Participants evaluated each haiku along seven impression dimensions: fluency, haiku-like wording, poeticness, coherence, understandability, favourability, and unexpectedness. Ratings were provided on a five-point Likert scale ranging from 1 (very low) to 5 (very high).

Across all models and seasonal splits, **impression ratings tend to concentrate in the mid-to-high range (values 3–4)**. This suggests that both AI-generated and human-written haiku are generally perceived as well-formed and readable, rather than polarising.

Among the evaluated dimensions, **fluency, haiku-like wording, and coherence** show the **strongest concentration toward higher ratings**. These dimensions consistently receive the highest scores across models, suggesting that most haiku, regardless of authorship,

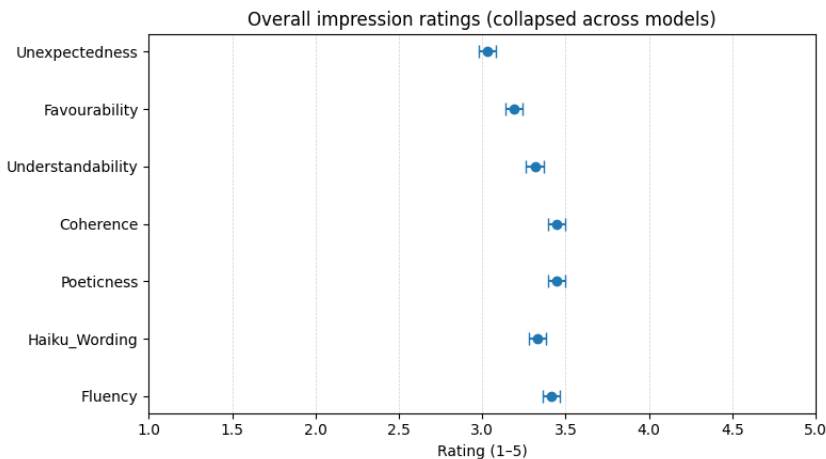


Figure 6.1. Surface Ratings (Global Pattern) - Estimated marginal means plot with 95% confidence intervals

are perceived as linguistically natural and structurally coherent.

In contrast, **unexpectedness tends to cluster around the midpoint of the scale**. This suggests that participants perceived most poems as moderately novel, rather than either strongly surprising or highly conventional. The **mid-range distribution of unexpectedness** indicates that novelty alone is unlikely to function as a reliable indicator of authorship.

Confidence ratings show a noticeably different pattern from impression ratings. For several model–season combinations, particularly StableLM Gamma-7B and LLM-JP-Instruct4 in the Autumn–Winter split, confidence values shift towards the lower end of the scale, even when impression ratings remain relatively high. This suggests that participants often judged the haiku as acceptable or aesthetically pleasing, yet remained uncertain as to whether the poem had been written by a human or generated by an AI system.

Model-specific patterns are also evident. Outputs produced by Gemma, Gemini and ChatGPT-5 show relatively consistent clustering around ratings of 3–4 across most impression dimensions. In contrast, StableLM Gamma-7B shows a clearer difference between impression ratings and confidence scores. This suggests that acceptable surface quality does not necessarily lead to confident source attribution.

Overall, the distribution of impression ratings suggests that AI-generated and human-written haiku are perceived as being broadly similar in terms of surface-level linguistic quality and stylistic plausibility.

6.2 Overall Recognition Accuracy

Participants were asked to judge whether each haiku had been written by a human or generated by an AI system. Recognition accuracy therefore reflects the proportion of correct authorship classifications across all responses.

Accuracy levels varied substantially across models. While several systems produced performance levels close to chance, others showed moderate detectability.

The closed-source models ChatGPT-5 and Gemini, and the open-source model StableLM Gamma-7B, produced recognition accuracies that were consistent with a classification based on chance. ChatGPT-5 achieved an accuracy rate of 49.3%, while Gemini reached 50.3% and Gamma-7B 52.4%. Participants were largely unable to reliably distinguish between AI-generated and human-written haiku.

In contrast, three models exhibited moderate levels of distinguishability. Gemma achieved an overall accuracy of 61.1%, LLM-JP-Instruct4 reached 59%, and LLaMA2 produced the highest recognition accuracy at 67%.

Although these values indicate some degree of detectability for certain models, overall performance remains relatively modest. While LLaMA2-generated haiku were the easiest for participants to distinguish from those written by humans, a substantial number of poems were still misclassified.

To better understand the sources of these errors, confusion matrices were examined for each model. These matrices distinguish between correct and incorrect classifications for each category (human and AI), revealing whether participants systematically favor one type of error over another.

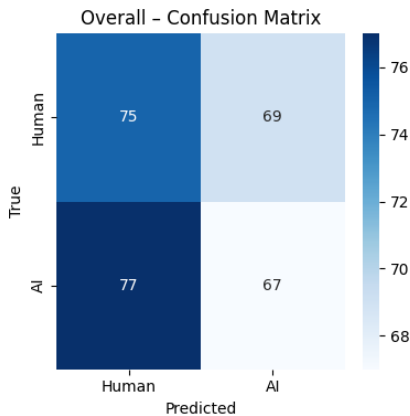


Figure 6.2. ChatGPT-5 Confusion Matrix

The confusion matrix for ChatGPT-5 reveals a noticeable asymmetry in error patterns. 77 AI-generated haiku were misclassified as human-written, while 69 human-written haiku were misclassified as AI-generated. However, correct classifications remain relatively balanced, with 75 human haiku and 67 AI haiku correctly identified.

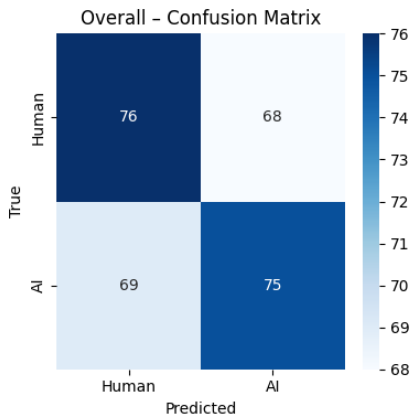


Figure 6.3. Gemma Confusion Matrix

StableLM Gemma-7B exhibits a more symmetrical error pattern, with 76 human and 75 AI haiku correctly classified, alongside 68 human and 69 AI misclassifications. This near symmetry suggests that participants found both classes equally challenging.

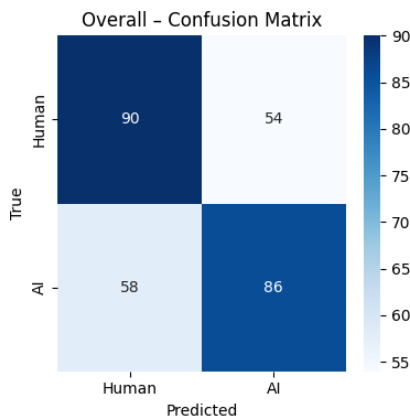


Figure 6.4. Gemma Confusion Matrix

The Gemma model achieved clearer separation, correctly classifying 90 human haiku and 86 AI haiku. However, misclassification remains common, suggesting that the outputs of the model still overlap significantly with those of humans.

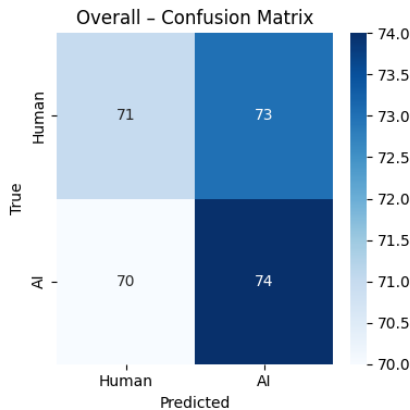


Figure 6.5. Gemini Confusion Matrix

Gemini exhibits an error pattern that is almost symmetrical. Of the 145 haiku analysed, 71 were correctly classified as human-written and 74 as AI-generated. However, 73 human-written haiku were misclassified as AI-generated and 70 AI-generated haiku were misclassified as human-written. These results are consistent with the near-chance overall accuracy, indicating that participants were largely unable to distinguish between Gemini outputs and human-written poems.

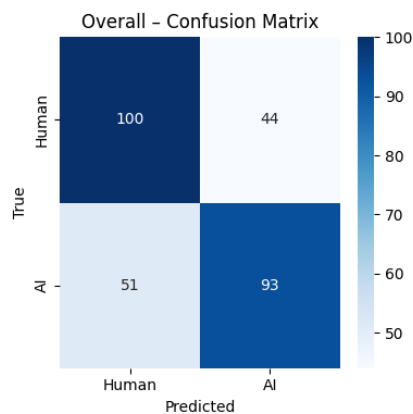


Figure 6.6. LLaMa Confusion Matrix

Overall, LLaMA2 shows the strongest separation, correctly identifying one hundred human haiku and classifying ninety-three AI haiku. However, despite this improvement, errors remain significant.

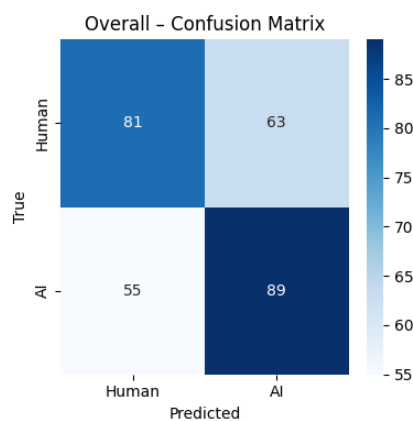


Figure 6.7. LLM-jp-instruct4 Confusion Matrix

Instruct4 also demonstrates moderate detectability, with eighty-one human and eighty-nine AI haiku correctly classified.

The confusion matrices across all models reveal substantial overlap between AI-generated and human-written haiku in terms of perceived cues. These results confirm that recognising authorship is far from trivial, and that misclassification patterns remain frequent in both directions even when moderate accuracy is observed.

6.3 Haiku-Level Accuracy and Uncertainty

In addition to model-level accuracy, recognition performance was also examined at individual poems level. Figure 6.8 shows a heatmap of the classification accuracy of each haiku, across all the models and seasonal splits that were evaluated.

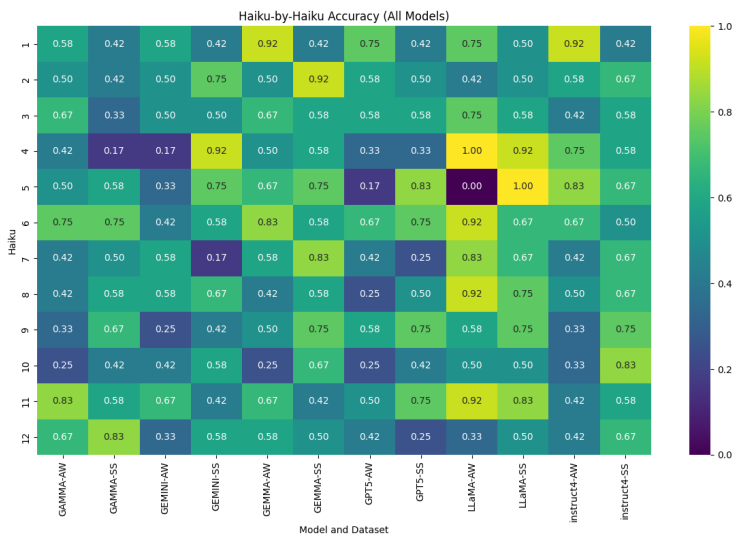


Figure 6.8. Haiku-by-haiku recognition accuracy across models and seasonal questionnaire splits.

The heatmap reveals substantial item-level variability in recognition performance. While some haiku are consistently classified correctly across respondents, others are frequently misclassified. This indicates that detectability is not uniform across the dataset and that specific poems may contain cues that make their authorship easier or harder to infer.

Several poems demonstrate relatively high recognition accuracy across multiple models. For instance, a few items achieve values above 0.80, approaching perfect recognition for certain systems. This suggests that respondents were able to identify the source of these poems with a high degree of consistency, possibly because some items contained cues that made authorship easier to infer. Conversely, some poems had very low accuracy values, sometimes approaching 0.20 or below. This suggests that respondents misclassified these poems systematically, indicating that some poems contained cues that made source attribution particularly challenging or misleading.

A second notable pattern relates to the differences between the models. Models such as LLaMA2 and Gemma exhibit a higher proportion of haiku with high accuracy, suggesting that participants found it easier to identify some of their outputs as AI-generated. By contrast, ChatGPT-5, Gemini and StableLM Gamma-7B exhibit more values clustered around the mid-range of the scale, reflecting the near-chance recognition accuracy observed in the overall model-level analysis.

The heatmap also enables comparison of the two seasonal questionnaire splits (autumn–winter and spring–summer). Although minor differences can be observed, no clear systematic pattern emerges between the two splits. Accuracy values appear to be distributed evenly across both seasonal conditions, suggesting that the seasonal content of the poems does not significantly impact authorship recognition.

Overall, the heatmap illustrates the varied nature of recognition performance at the item level. Rather than being uniformly easy or difficult to classify, individual haiku differ substantially in terms of how easily their authorship can be inferred.

To further examine the reliability of these differences at item level, per-haiku accuracy estimates were visualized using bar plots with 95% confidence intervals. These plots show that many haiku have wide confidence intervals, suggesting significant uncertainty in the estimated recognition rate.

Figures 6.9–6.14 present the per-haiku recognition accuracy for each model, together with the 95% confidence intervals, separately for the Autumn–Winter (AW) and Spring–

Summer (SS) questionnaire splits.

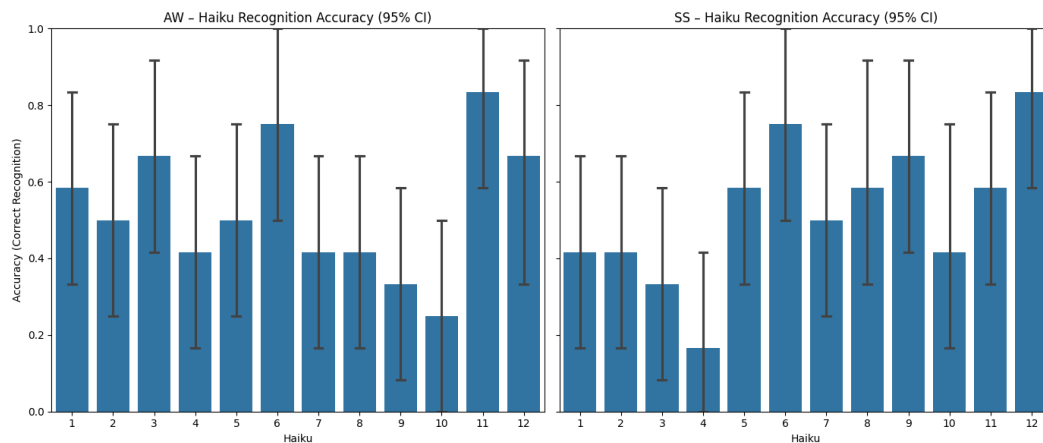


Figure 6.9. Per-haiku recognition accuracy for StableLM Gamma-7B with 95% confidence intervals (AW and SS splits).

For the StableLM Gamma-7B model, most haiku have accuracy values close to the mid-point of the scale and the corresponding confidence intervals often overlap the chance-level threshold of 0.5. This suggests that participants were generally unable to reliably distinguish between the model's outputs and human-written haiku. Only a few poems have noticeably higher or lower recognition rates, and even these often have relatively wide confidence intervals.

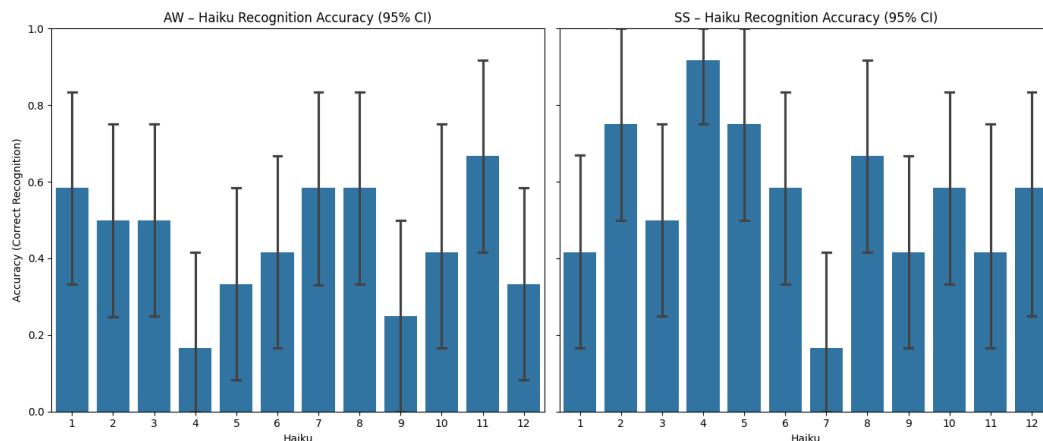


Figure 6.10. Per-haiku recognition accuracy for Gemini with 95% confidence intervals (AW and SS splits).

A similar pattern is observed for Gemini. Recognition accuracy remains widely distributed around chance level across individual haiku, with many overlapping confidence intervals. Although a few poems achieve moderately higher accuracy values, most remain difficult for participants to classify reliably, which confirms the near-random accuracy observed in the aggregated results.

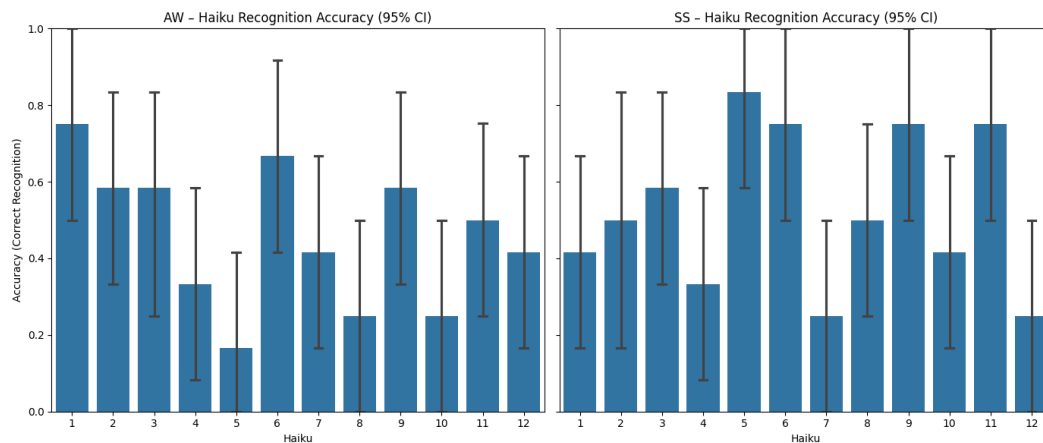


Figure 6.11. Per-haiku recognition accuracy for ChatGPT-5 with 95% confidence intervals (AW and SS splits).

While a few haiku achieve relatively high recognition accuracy, most items fall within the mid-range of the scale and exhibit wide confidence intervals. This pattern suggests that participants were largely uncertain about attributing authorship to ChatGPT-5-generated poems.

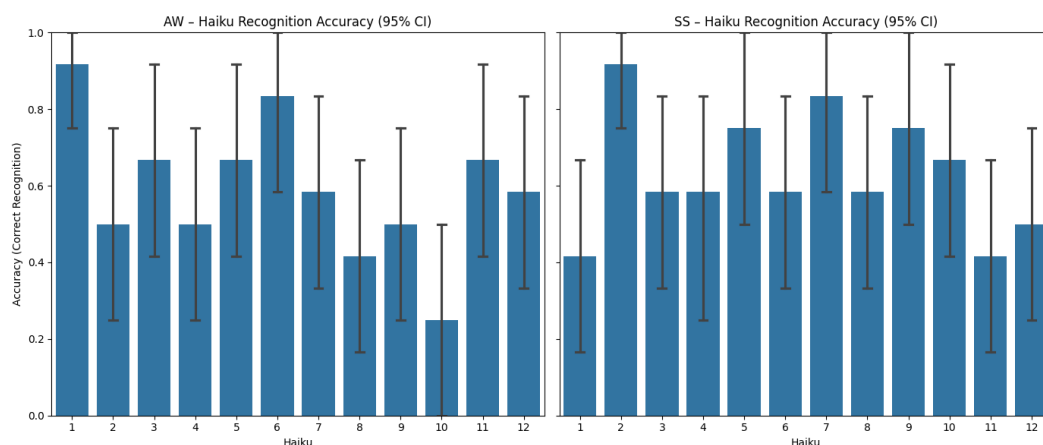


Figure 6.12. Per-haiku recognition accuracy for Gemma with 95% confidence intervals (AW and SS splits).

Gemma's recognition performance varies slightly more across poems. Some items achieve high accuracy values, suggesting that certain generated haiku contain features that participants interpret as indicative of machine authorship. However, many items still show confidence intervals that overlap with the chance level, indicating that these cues are not consistently present in all outputs.

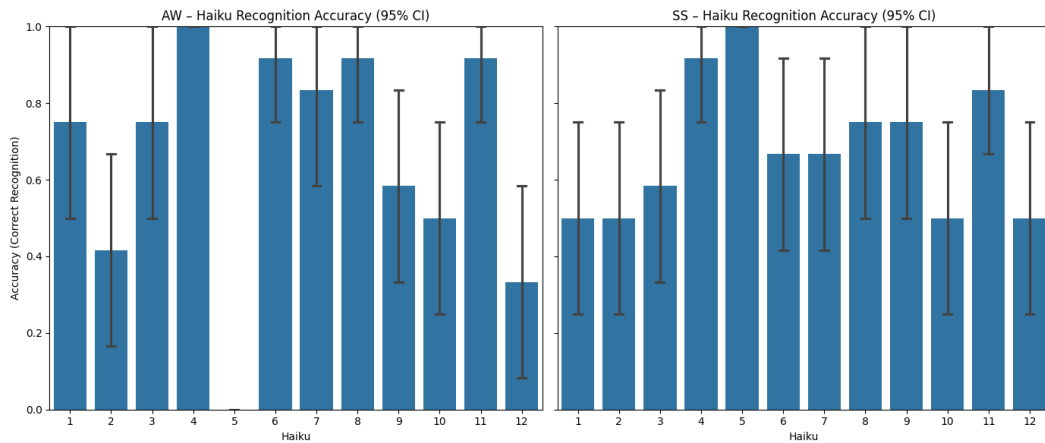


Figure 6.13. Per-haiku recognition accuracy for LLaMA2 with 95% confidence intervals (AW and SS splits).

Compared with previous models, LLaMA2 shows a wider spread of accuracy values across poems. Several haiku achieve relatively high recognition accuracy, particularly in the spring–summer split, which suggests that some of the generated poems contain noticeable stylistic patterns. Nevertheless, even for this model, many items are associated with large confidence intervals, indicating substantial uncertainty in recognition performance.

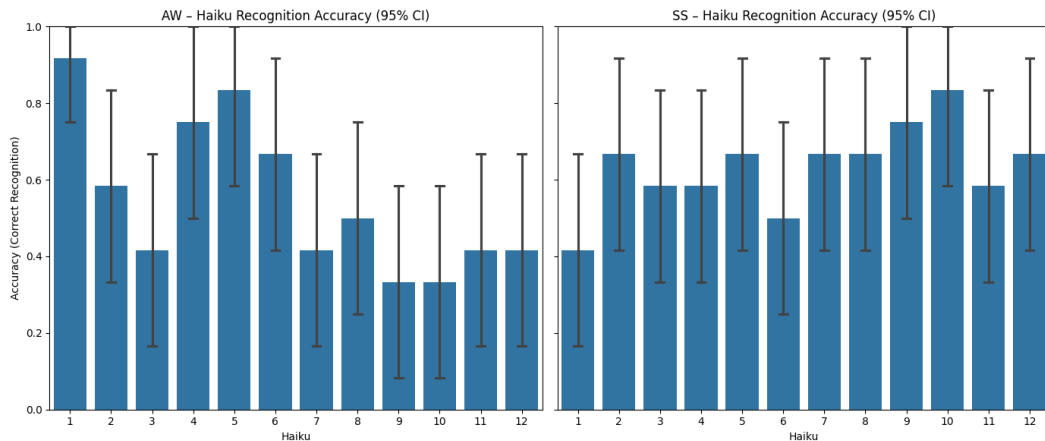


Figure 6.14. Per-haiku recognition accuracy for LLM-JP-Instruct4 with 95% confidence intervals (AW and SS splits).

Finally, the LLM-JP-Instruct4 model exhibits moderate variability across poems. While some items exhibit relatively high recognition rates, a large proportion of poems cluster around mid-range accuracy values with overlapping confidence intervals. As with the other models, this suggests that authorship cues are inconsistent and unevenly distributed across individual haiku.

Confidence interval analysis across all models confirms that recognition performance varies substantially across individual poems. Only a limited number of poems exhibit clearly distinguishable recognition patterns, while the majority remain difficult to classify reliably. This supports the idea that recognising authorship depends strongly on the specific characteristics of individual poems, rather than systematic stylistic differences between human-written and AI-generated haiku.

6.4 Item Dependence and Detectability

The descriptive analyses reveal a high degree of dependence between items in terms of recognition performance. Although there is variation in overall model-level accuracy across systems, this is primarily driven by a small subset of highly diagnostic haiku.

While most poems cluster around chance-level classification, while only a few items consistently achieve high or low recognition accuracy. This pattern suggests that detectability is not determined by a stable global signal across all poems, but rather by specific item-level cues.

When considered as a whole, the stacked bar plots, confusion matrices, heat maps and per-haiku accuracy plots all point to the same conclusion: participants' judgements are heavily influenced by surface-level impressions and item-specific features.

Although some models produce outputs that are moderately distinguishable from human-written haiku, reliable discrimination remains limited across the entire dataset. Therefore, recognition performance appears to depend primarily on the characteristics of individual poems rather than on consistent differences between human and AI authorship.

Chapter 7

Inferential Analysis

While the previous chapter presented descriptive patterns observed in the data, this chapter examines whether these patterns remain statistically significant when participant-level variability, item-level variability, and subjective impression ratings are considered. The analysis addresses three related questions in particular. First, it considers whether participants' background characteristics influence their ability to distinguish AI-generated haiku from human-written ones. Second, it considers whether the impression ratings provided by the participants contain cues that predict correct authorship recognition. Third, it considers whether the cues that make a poem appear to be written by a human or an AI correspond to the actual source of the poem.

To address these questions, a combination of logistic regression and Bayesian generalised linear mixed models (GLMMs) was employed. The mixed-effects framework takes into account the hierarchical structure of the dataset, in which each respondent evaluated multiple poems, and each poem was evaluated by multiple respondents.

7.1 Participant Background and Recognition Ability

The first inferential analysis investigated whether participants' background characteristics affected their ability to correctly identify haiku that were either AI-generated or written by humans.

To address this question, a per-respondent logistic regression model was estimated. The dependent variable indicates whether a participant achieved above-chance recognition accuracy, defined as correctly classifying more than half of the haiku they evaluated. The predictor variables were Japanese language proficiency, academic background (university department), frequency of haiku reading and experience of composing haiku.

Because the open-ended responses describing academic departments were highly heterogeneous, they were aggregated into broader disciplinary categories in order to reduce sparsity and improve interpretability. These categories included *STEM*, *Social Sciences and Humanities*, *Psychology*, and *Unknown or unspecified departments*. Model coefficients were interpreted using odds ratios (OR), where values greater than one indicate increased odds of above-chance recognition and values below one indicate reduced odds.

Feature	Coef	Odds Ratio
Japanese_proficiency_N1	-0.778	0.459
Grade_group_STEM	0.567	1.764
Haiku_composition_Rarely	-0.441	0.643
Japanese_proficiency_N2	-0.393	0.675
Haiku_experience_A few/month	0.354	1.425
Japanese_proficiency_None/Unknown	-0.351	0.704
Japanese_proficiency_N5	-0.176	0.838
Haiku_composition_Occasionally	0.153	1.165
Grade_group_Unknown/No department	0.138	1.148
Haiku_experience_Occasionally	0.120	1.128
Japanese_proficiency_N4	-0.062	0.939
Haiku_composition_Sometimes	-0.061	0.941
Haiku_experience_1/2 per week	-0.042	0.959
Grade_group_Psychology	0.014	1.014

Table 7.1. Logistic regression coefficients and corresponding odds ratios for features influencing above-chance recognition.

The coefficient estimates suggest that Japanese language proficiency, relative to that of native speakers, is most strongly associated with recognition performance. Several levels of non-native proficiency display lower odds of achieving above-chance classification accuracy, suggesting that subtle linguistic cues in the haiku may influence the attribution task.

By contrast, the frequency with which participants read haiku and their experience of composing haiku show weak and inconsistent relationships with recognition accuracy. Participants who reported frequent exposure to haiku did not systematically outperform those with less experience. This indicates that familiarity with the poetic form alone does not provide reliable cues for distinguishing between AI-generated and human-written poems.

An unexpected pattern emerges for participants from STEM backgrounds: they show slightly higher odds of above-chance recognition than participants from the social sciences and humanities. However, this result should be interpreted with caution, as it may be due to differences in response strategies, familiarity with AI-generated text or sampling variability, rather than representing a meaningful disciplinary effect.

To assess the robustness of these findings, two additional models were estimated. The first model included Japanese language proficiency as the only predictor, while the second model included all background variables except language proficiency.

Table 7.2 reports the results for the language-only model.

Feature	Coef	Odds Ratio
Japanese_proficiency_N1	-0.690	0.501
Japanese_proficiency_N2	-0.411	0.663
Japanese_proficiency_None/Unknown	-0.214	0.808
Japanese_proficiency_N5	-0.025	0.975
Japanese_proficiency_N4	-0.017	0.984

Table 7.2. Logistic regression results for the language-only model.

Table 7.3 reports the results for the model excluding language proficiency.

Feature	Coef	Odds Ratio
Grade_group_STEM	0.496	1.642
Haiku_experience_A few/month	0.421	1.524
Haiku_composition_Rarely	-0.414	0.661
Haiku_experience_Occasionally	0.240	1.271
Haiku_composition_Occasionally	0.181	1.198
Grade_group_Unknown/No department	0.119	1.126
Haiku_composition_Sometimes	-0.049	0.952
Haiku_experience_1/2 per week	0.010	1.010
Grade_group_Psychology	0.010	1.010

Table 7.3. Logistic regression results for the model excluding language proficiency.

The predictive performance was evaluated using a three-fold cross-validated Area Under the Curve (AUC), which provides an estimate of the classification performance based on data not used for training. This approach was chosen because the small sample size meant that it was not possible to reliably estimate both the model parameters and the performance of the test set using a conventional training–testing split.

Across all models, AUC values remained close to chance:

- Full model (all predictors): AUC (CV 3-fold) = 0.416
- Language-only model: AUC (CV 3-fold) = 0.456
- No-language model: AUC (CV 3-fold) = 0.432

These results suggest that participants’ background characteristics offer little predictive value in distinguishing above-chance recognition ability. Although language proficiency appears to contain some useful information, the overall predictive performance remains poor. This suggests that recognition performance is more strongly dependent on the characteristics of individual poems than on stable participant-level traits.

7.2 Impression Cues and Correct Recognition

The second analysis investigated whether subjective impression ratings could help participants correctly identify the true author of a haiku.

To investigate this question, a Bayesian generalised linear mixed model was used to analyse whether participants correctly classified the authorship of poems. The predictor variables included the seven impression ratings (fluency, haiku-like wording, poeticness, coherence, understandability, favourability, and unexpectedness), as well as perceived thematic relevance.

A mixed-effects framework was employed because the dataset had a crossed hierarchical structure, in which each respondent evaluated multiple poems, and each poem was evaluated by multiple respondents. Random intercepts were therefore included for both respondents and haiku to account for individual differences in judgement behaviour, as well as variability between poems.

Feature	Coef	SD	Odds Ratio
Favourability	-0.189	0.014	0.828
Understandability	0.081	0.014	1.084
Unexpectedness	0.077	0.015	1.080
Theme_Relevance	0.075	0.013	1.078
Fluency	0.053	0.014	1.054
Poeticness	-0.042	0.013	0.959
Coherence	-0.018	0.014	0.983
Haiku_Wording	-0.009	0.014	0.991

Table 7.4. Mixed-effects model estimates for impression cues predicting correct recognition.

The results from Table 7.4 suggest that impression ratings offer only limited predictive value in terms of correct recognition. Although some rating dimensions show weak associations with correct classification, these effects are small and inconsistent. In other words, the subjective qualities that make a haiku appear aesthetically pleasing or coherent do not reliably indicate authorship. Participants often rate poems positively, regardless of whether they were generated by humans or language models.

7.3 What Makes a Haiku Feel Human

The third analysis focused on perceived authorship rather than objective recognition accuracy. Specifically, a Bayesian mixed-effects logistic regression model was used to estimate the probability of a poem being judged as human-written.

Table 7.5. Posterior coefficients, standard deviations, and odds ratios for impression cues predicting correct recognition

Feature	Coef	SD	Odds Ratio
Favourability	0.539466	0.016896	1.715091
Haiku_Wording	0.309418	0.016220	1.362631
Theme_Relevance	0.286363	0.014761	1.331576
Unexpectedness	0.201238	0.017786	1.222916
Coherence	0.141988	0.015673	1.152563
Fluency	0.112324	0.015777	1.118875
Understandability	0.072880	0.016130	1.075601
Poeticness	-0.030810	0.015581	0.969660

The results suggest that poems judged as human-written are more likely to receive higher ratings for poeticness, coherence and haiku-like wording. Therefore, poems that are perceived as stylistically consistent with traditional haiku conventions are more likely to be attributed to human authors.

Imagery richness and perceived poetic quality appear to play a central role in human attribution judgements. Participants tend to interpret evocative or emotionally resonant qualities in a poem as signals of human creativity. However, these cues do not necessarily correspond to the poem's actual origin. Some AI-generated haiku receive high ratings in this respect and are therefore often mistakenly classified as having been written by a human.

7.4 What Makes a Haiku Feel AI

A complementary model examined the cues that lead participants to judge a poem as AI-generated.

Table 7.6. Posterior coefficients, standard deviations, and odds ratios for impression cues predicting classification as AI-generated

Feature	Coef	SD	Odds Ratio
Favourability	-0.539480	0.016887	0.583051
Haiku_Wording	-0.309397	0.016213	0.733889
Theme_Relevance	-0.286554	0.014757	0.750847
Unexpectedness	-0.201124	0.017781	0.817811
Coherence	-0.142124	0.015667	0.867514
Fluency	-0.112353	0.015772	0.893728
Understandability	-0.072781	0.016124	0.929804
Poeticness	0.030815	0.015576	1.031295

This analysis shows that lower ratings for poetic quality and stylistic authenticity increase the probability that a poem is machine-generated. Those perceived as formulaic, semantically shallow or lacking emotional resonance are more likely to be attributed to AI.

However, the strength of these associations is modest. Many poems that receive mid-range ratings are still difficult for participants to classify with confidence.

This suggests that AI-generated haiku often occupy the same perceptual space as poems written by humans, particularly when they satisfy basic structural and stylistic expectations.

7.5 Perceived Authorship vs Ground Truth

To further clarify the relationship between perceived and actual authorship, the impression ratings were grouped in two complementary ways. First, ratings were grouped by the *true source* of the haiku: S_AI for genuinely AI-generated poems and S_Human for genuinely human-written poems. Second, ratings were grouped by the *perceived source*: T_AI for poems judged by participants as AI-generated and T_Human for poems judged as human-written, regardless of their actual origin.

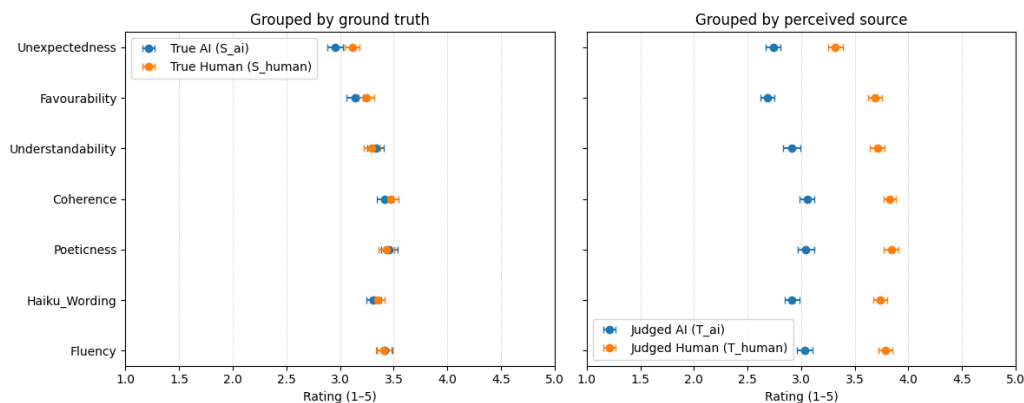


Figure 7.1. Impression ratings grouped by ground truth (left) and by perceived authorship (right). S_AI and S_Human denote ratings grouped by true source, whereas T_AI and T_Human denote ratings grouped by participants' judgments. Error bars represent 95% confidence intervals.

Figure 7.1 clearly shows the difference between these two comparisons. When ratings are grouped by ground truth, the distributions of S_AI and S_Human overlap substantially across all impression dimensions. This suggests that AI-generated and human-written haiku are not consistently perceived as qualitatively distinct based on their true authorship alone.

By contrast, when ratings are grouped by perceived source, the separation becomes much more pronounced. Across all dimensions, poems judged as human-written (T_Human) receive consistently higher ratings than poems judged as AI-generated (T_AI). This pattern is particularly evident in terms of favourability, haiku-like wording, coherence, fluency, and understandability. Therefore, the figure suggests that impression ratings are more strongly associated with perceived humanness than with the poems' actual authorship.

This comparison reveals a discrepancy between perception and accuracy. The cues that create the impression of authorship are not strongly associated with correct classification. Participants therefore rely on heuristics to interpret a poem's origin, but these heuristics are not consistently reliable.

For example, poems that are rated as highly poetic are often judged to have been written by a human, even when they were actually generated by AI. Conversely, poems that are perceived as less expressive are often attributed to AI, even when they were written by humans. These patterns suggest that participants rely on their cultural expectations of creativity and poetic style rather than objective indicators of authorship.

7.6 Cross-analysis of Recognition and Perceived Humanness

Taken together, the inferential analyses reveal an important distinction between recognition accuracy and perceived humanness.

Recognition accuracy is measured by how well participants can correctly identify the true author of a poem. Perceived humanness, on the other hand, reflects the subjective cues that lead participants to believe a poem was written by a human. The results suggest that these two processes are only weakly related.

Participants' judgements are primarily influenced by surface-level aesthetic impressions rather than cues that can reliably distinguish between AI-generated and human-written text. This finding aligns with the descriptive results reported in the previous chapter, where recognition performance remained close to chance for several models, and classification accuracy varied significantly across individual poems.

Overall, the inferential analysis suggests that the attribution of authorship in poetry is primarily influenced by perception rather than stable stylistic differences between human and AI authors. As language models improve, their outputs increasingly resemble human-written poems aesthetically, making it progressively more difficult to discriminate reliably between them.

Chapter 8

Qualitative Evidence

While previous chapters examined the questionnaire data using descriptive and inferential statistical analyses, this chapter focuses on the open-ended responses provided by participants. The aim is to gain a better understanding of how respondents justified their judgements about authorship and the interpretive strategies they relied on to distinguish between AI-generated and human-written haiku. In this sense, the qualitative evidence supplements the quantitative analyses by making the reasoning of the participants more explicit.

This chapter analyses the optional written responses included in the questionnaire. These responses covered three main areas: **the cues participants used to distinguish between AI and human authorship**, **short comments on individual haiku**, and **reflections on poems or dimensions that were difficult to evaluate**. Responses were provided in both Japanese and English. For the purposes of interpretation, the comments were read iteratively and grouped into recurring themes using qualitative thematic analysis. Rather than treating each answer as an isolated judgment, the analysis identifies the main interpretive tendencies that emerged repeatedly across respondents.

Overall, the responses suggest that participants did not consider a wide range of highly specific criteria. Instead, they repeatedly referred to a relatively limited set of cues, such as adherence to formal rules, clarity of meaning, the use of imagery, emotional resonance, the naturalness of the wording and assumptions about what AI-generated language should sound like. At the same time, the responses reveal important differences between participants who were more familiar with haiku and those who relied more on their initial impressions or intuition.

8.1 Recurring Attribution Heuristics

The open-ended responses reveal a clear pattern: participants repeatedly relied on a small number of recurring heuristics when attributing haiku to human or AI authorship. While these heuristics do not necessarily correspond to objective indicators of authorship, they represent the cues that respondents considered relevant when making their judgments.

One major category consists of **formal and rule-based cues**. Many participants explicitly referred to traditional haiku conventions, such as the use of seasonal words (*kigo*), the expected line structure and natural word order and phrasing. Several respondents stated directly that they looked at "how haiku is related to the theme" or at "the use of *kigo* and whether the scene came to mind." Others referred more broadly to whether the wording sounded like haiku-specific language, for example, whether the poem used traditional or poetic expressions. Japanese participants sometimes commented that a poem felt unnatural due to an unusual orthographic form or because a certain expression was something "a human would never write." These comments suggest that formal conformity was often con-

sidered a sign of humanity, even when the judgement was based more on impression than technical verification.

A second recurring heuristic concerns **surface-level textual features associated with AI**. Participants often described poems labelled as AI-generated as repetitive, overly perfect, direct to the point of being flat, or mechanically assembled. One English-language respondent wrote that a poem seemed AI-generated because it was "too perfect, not unique", while another commented that it was "too straight, too clear in the meaning." Japanese responses expressed similar ideas, with comments suggesting that certain vocabulary felt commonplace, overly general or lacking in individuality. One participant remarked that poems with universality but no particular *peculiarity* (癖) or distinctive character seemed more AI-like. Another respondent described some texts as feeling as though words had simply been combined rather than poetically composed. These judgments reflect common expectations about machine-generated text. AI is imagined as producing language that is structurally plausible but compositionally shallow.

A third major heuristic concerns **clarity of meaning and coherence**. Many respondents described poems that were judged to be AI-generated as being difficult to understand, unclear in their intention or disjointed. Several comments are very direct: "I don't understand it", "I don't get the first line", or "I can't understand what it is trying to say". Similarly, in Japanese, some respondents similarly wrote that they did not understand why a poem evoked autumn, or that the semantic connection between the elements was unclear. By contrast, poems judged as human-written were often described as easy to follow, coherent or successful in communicating their message. One respondent stated that they judged a haiku based on whether "what the poem intends to say is easy to understand", while another simply wrote that it was "easy to understand" and therefore convincing. This suggests that comprehensibility is a strong indicator of perceived human authorship. Respondents were more likely to attribute a poem to a human author when they felt that the scene or meaning could be grasped without effort.

A fourth recurring heuristic concerns **imagery and emotional engagement**. Haiku that were judged to be written by humans were frequently described in terms of their ability to evoke a visual scene, generate atmosphere or create emotional resonance. One respondent wrote that they chose human authorship when "an image appeared in my mind", while another commented that the poem "paints a scene in my head". Japanese respondents often referred to whether a haiku vividly depicted a *scene* (情景) or allowed a particular atmosphere to be imagined. Some also associated emotional subtlety with human authorship, for example suggesting that a poem focused on a felt moment rather than merely stating an idea. These comments suggest that readers associate poetic vividness and emotional resonance with human creativity. However, as the quantitative chapters have shown, such cues were not necessarily reliable indicators of true authorship. Above all, they functioned as cues of *perceived humanness*.

A fifth related heuristic concerns **specificity versus obviousness**. Several respondents seemed to prefer poems that focused on one precise and evocative idea, while distrusting those that stated too much too explicitly. One English respondent wrote that human poems focus more on "emotions and abstract elements that express a feeling or a moment", whereas some AI-like poems were described as overly explanatory. Similarly, some Japanese respondents interpreted the explicit inclusion of multiple related symbols as being less human-like, suggesting that a human poet would imply rather than overstate. This reflects the broader expectation that successful haiku should be suggestive rather than merely descriptive.

Taken together, these recurring heuristics reveal that participants approached the task using relatively consistent internal criteria. They repeatedly searched for formal regularity, semantic clarity, evocative imagery, emotional depth and signs of natural poetic composition. However, these criteria primarily functioned as *perceptual heuristics*, rather than as

consistently valid indicators of ground truth. This helps to explain why respondents could apply meaningful judgement strategies yet still produce recognition patterns close to chance for several of the strongest models.

8.2 Knowledge-Based vs Intuition-Based Reasoning

A second important distinction that emerges from the open-ended responses concerns the difference between **knowledge-based reasoning** and **intuition-based reasoning**. While these two modes are not entirely distinct, they reflect the two different approaches that participants took to the authorship attribution task.

Some participants displayed a relatively explicit and knowledge-based approach. They referred to recognised features of haiku and Japanese poetic tradition in their comments, including seasonal vocabulary, classical particles, rhythm, orthographic choices and culturally marked forms of expression. For instance, one respondent commented specifically on the use of the archaic spelling ” 𐰇 ” instead of the more common modern form ” え ”, noting that this created a sense of strangeness while the poem still felt complete as a haiku. Another respondent discussed whether the language resembled haiku-like diction, citing examples such as 〰 けり . Others reflected on the appropriateness of the seasonal imagery used and whether it sounded like something a human poet would intentionally choose. These responses suggest an evaluation method based on some prior familiarity with literary conventions.

This knowledge-based approach is also evident in comments that tolerate ambiguity. Respondents with a greater sensitivity to haiku did not always view obscurity as a failure. In some cases, when a poem was difficult to interpret, they were willing to consider the difficulty as a sign of poetic subtlety rather than evidence of machine generation. This is particularly evident in responses to difficult items, where ambiguity was sometimes described as meaningful or compatible with human authorship.

By contrast, many other responses were guided more strongly by intuition. These comments focused on whether the poem felt natural and made immediate sense, whether its tone seemed human, and whether the respondent could visualise the scene easily. Several answers are notably brief and impressionistic: ”general feel, mostly”, ”originality”, or simply ”I don’t understand it”. Some respondents explicitly stated that they were not familiar with haiku and therefore had no clear basis for comparison. Others mentioned that they relied on the overall flow of the words, their placement, or whether the poem seemed to contain ”human sense”. Rather than applying explicit literary criteria, they relied on their overall impression of the poem’s tone, atmosphere and intelligibility.

This distinction is important because it helps to explain some of the variability observed in the dataset. The same textual property may be interpreted differently depending on the respondent’s background and reasoning style. For instance, ambiguity can be evaluated in two ways that are almost opposite to each other. A participant with a good knowledge of haiku might view ambiguity as suggestive, allusive or aesthetically meaningful. However, a participant who relies mainly on immediate intelligibility might interpret the same ambiguity as confusion, weak coherence or machine-like opacity.

A similar contrast emerges in responses referring to **traditional versus modern atmosphere**. Some participants seemed to associate ”traditional” sounding haiku with human authorship, while more unusual or modern-seeming phrasing was associated with AI authorship. One respondent even remarked that some items ”looked like French poem”, indicating a sense that the poem departed from the expected atmosphere of haiku. Others explicitly linked human authorship to subtle, indirect or old-fashioned poetic texture. This suggests that judgments were shaped not only by the text’s intrinsic qualities, but also by culturally informed expectations of what a human-written haiku should sound like.

Overall, the open-ended data suggest that participants did not apply a single, unified decision rule. Instead, they used a variety of reasoning methods, ranging from explicit literary knowledge to impressionistic intuition. This heterogeneity is important when interpreting the broader results of the study because it suggests that attribution judgments did not stem from a single, shared evaluative standard.

8.3 Instability of Unexpectedness

The open-ended responses also provide valuable insight into one of the questionnaire's most unstable dimensions **Unexpectedness**. As the descriptive and inferential analyses showed, this variable was not a strong or consistent predictor of correct recognition. The qualitative material helps to clarify this.

Many participants found unexpectedness difficult to evaluate, but this was not simply because some poems were more surprising than others. Rather, the responses suggest that participants did not share a common understanding of what 'unexpectedness' should mean in the context of haiku. Several comments explicitly mention this uncertainty. One respondent wrote that it was difficult to know how to judge whether a haiku was unexpected, while another explained that they did not know what should or should not count as unexpected in haiku. An English-language participant asked whether unexpectedness referred to surprise at the content or idea or to strange phrasing alone. These responses demonstrate that the concept itself was interpreted in various ways.

For some respondents, unexpectedness seemed to refer to the presence of an imaginative twist, striking juxtaposition or originality. In such cases, the absence of an obvious poetic twist could be interpreted as a sign of weak creativity, and thus a possible indication of AI-generated content. However, for other respondents, a haiku did not need to be strikingly surprising to feel human. Instead, subtlety, quietness and understatement could also be interpreted as signs of poetic quality. In these instances, a poem that appeared overly clever, visibly constructed or strongly marked by novelty could itself feel artificial.

Another related source of instability concerns **ambiguity**. Several respondents identified poems that, while grammatically correct, were difficult to interpret, as being particularly challenging to rate. One English response noted that some items could be understood "in both ways", either as something deeper or simply as a natural image. Another respondent wrote that the difficult items were those that lacked uniqueness, yet still did not feel clearly AI-generated, as they retained "some kind of context". These comments suggest that poems often occupy an uncertain space between poetic indirection and semantic vagueness. For some participants, this ambiguity suggested depth, while for others it suggested a failure of communication.

The responses further show that **fluency alone was not sufficient to resolve this uncertainty**. Some items were described as easy to read but lacking depth or a strong idea. Others were clear on the surface, yet difficult to classify in terms of authorship. This distinction is important: a poem may be grammatically smooth and structurally plausible yet still fail to provide a clear indication of whether it is human or artificial. In other words, fluency and unexpectedness did not converge to form a single evaluative dimension.

Unexpectedness was also affected by differences in cultural and linguistic competence. Some respondents wrote that haiku involving Japanese-specific terms, poetic diction or cultural references were more challenging to evaluate, as they lacked the confidence to judge the grammar or nuances accurately. This suggests that the perception of surprise or originality is inextricably linked to the reader's own familiarity with the form and language.

Overall, the instability of unexpectedness highlights a wider issue in the evaluation of poetry. Some of the features that contribute to poetic richness, such as indirection, subtlety, ambiguity and suggestive incompleteness, also make it more difficult to judge authorship.

In the present study, unexpectedness did not function as a shared diagnostic cue because participants attributed different meanings to it and interpreted its presence or absence through different evaluative frameworks.

8.4 Convergence with Quantitative Findings

The qualitative evidence strongly supports the patterns reported in the descriptive and inferential chapters. In particular, it helps to explain why participants' judgments were often internally consistent yet only loosely connected to the poems' actual source.

First, the open-ended responses support the finding that **perceived humanness is more strongly tied to impression than to ground truth**. Participants repeatedly described human-labelled haiku as coherent, vivid, emotionally resonant and stylistically natural. Conversely, AI-labelled haiku were often described as repetitive, contrived, shallow, overly direct or unclear. This aligns with the mixed-effects results indicating that impression ratings were strongly associated with *judged* authorship, but only weakly associated with *correct* recognition. In other words, respondents were not choosing at random; they were applying meaningful internal criteria. However, these criteria were more effective at shaping the perceived source than identifying the actual source.

Second, qualitative responses help to explain the almost random performance observed in models such as ChatGPT-5, Gemini and StableLM Gamma-7B. If participants primarily consider factors such as coherence, readability, imagery, emotional engagement and atmosphere when evaluating a model's performance, then any model capable of producing outputs that satisfy these expectations can easily be perceived as human. This is consistent with confusion matrices and haiku-level analyses showing substantial overlap between machine-generated and human-written poems in terms of perceived cues. A poem that "paints a scene" or evokes an emotional response can be judged as human regardless of its true origin.

Third, the qualitative findings reinforce the interpretation that **recognition was strongly item-dependent**. Participants' comments often focused on the specific features of individual poems, such as an awkward break, a forced seasonal word, a vivid image or an unnatural choice of words. This is consistent with the quantitative result that detectability varied substantially from one haiku to another and that model-level accuracy was often driven by a limited subset of items rather than a consistent overall stylistic signal. The open-ended responses suggest that attribution frequently depended on small local cues rather than global stylistic consistency.

Fourth, the responses provide further evidence in support of the interpretation of a **halo-like attribution bias**. Many comments imply that participants associated poetic quality itself with human authorship. When a haiku was deemed aesthetically successful, emotionally engaging or culturally convincing, respondents often treated these qualities as evidence that it must have been written by a human. Conversely, when a poem felt flat, overly explicit or mechanically assembled, this was often taken as evidence of AI generation. This mirrors the inferential result that favourability, haiku-like wording, coherence and related impression variables strongly predicted perceived authorship. Therefore, the qualitative evidence clarifies the mechanism behind the statistical pattern: participants inferred the source based on the perceived poetic value.

Fifth, the open-ended responses help to explain why participant background variables had only a weak predictive value in the inferential models, despite there being clear differences in reasoning style. While some respondents demonstrated greater literary knowledge, others relied mainly on intuition; however, both groups often used cues that were only partially diagnostic of the actual source. Having greater knowledge changed how ambiguity or tradition was interpreted, but it did not necessarily lead to higher recognition accuracy. This

is consistent with the weak predictive performance of the participant-level logistic models.

Taken together, the qualitative responses suggest that the difficulty of the authorship attribution task does not stem from an absence of evaluative cues altogether. Rather, it is due to the fact that the cues participants rely on are only partially diagnostic. While they are meaningful as interpretations of poetic quality, they do not reliably distinguish between human and AI authorship. Consequently, participants can apply consistent and coherent judgment strategies while achieving low or only moderate recognition accuracy.

Overall, the qualitative evidence reinforces the key finding of this thesis: that contemporary haiku-generating language models can successfully occupy the perceptual space associated with human poetry. Readers rely on culturally and aesthetically meaningful heuristics when making attribution judgments, but these heuristics are not sufficient to guarantee correct recognition. This confirms that the evaluation of AI creativity cannot be reduced to formal correctness alone and that the perception of humanity is shaped as much by interpretive expectation as by the text's true origin.

Chapter 9

Human vs Machine: Grok Comparison

In addition to the human evaluation presented in previous chapters, an exploratory comparison was conducted using Grok as an automated external judge. Unlike a separate pilot study involving unrelated texts, this comparison was performed using the same set of haiku that were used in the questionnaire study. This analysis aimed to examine whether a large language model acting as a classifier would rely on similar cues to those used by human respondents and whether its behaviour would vary across model tiers when evaluated on the same authorship attribution task.

Grok was tasked with determining whether each haiku had been written by a human or an AI system, and with providing a textual justification for each decision. This procedure was carried out in two stages as part of the broader experimentation process. In the first version, multiple haiku were evaluated together within the same conversational context. A second version, emphasized in the present chapter, involved submitting each haiku individually, one at a time. The latter setting is more useful for analysis because it reduces the effects of contextual carryover between items, making the classification decision more directly attributable to the poem itself.

The evaluation set included the human-written haiku used in the questionnaire, alongside the AI-generated haiku produced by the six models considered in this thesis: ChatGPT-5, Gemini 2.5 Pro, StableLM Gamma-7B, Gemma-2b-jpn, LLaMA2 and LLM-JP-Instruct4. As Grok was applied to the same set of haiku used in the human questionnaire, any differences between Grok's responses and those of the human respondents cannot be attributed to changes in the evaluation material. Therefore, the comparison isolates differences in judging behaviour rather than differences in the underlying poems.

To facilitate comparison between the two Grok settings, Tables 9.1 and 9.2 summarize the classification results across models. The key measure is the proportion of AI-generated haiku that are correctly identified as such, alongside Grok's performance on the human-written baseline. This comparison reveals a significant asymmetry between model groups and moderate sensitivity to evaluation context. In both settings, Grok consistently classified human-written haiku as such and failed to detect AI-generated haiku from ChatGPT-5, Gemini and StableLM Gamma-7B, classifying all of them as human. However, smaller or less optimised models behaved differently in the two conditions. In the all-at-once setting, Grok perfectly separated Gemma, LLaMA2 and Instruct4 from the human baseline. In the one-by-one setting, however, this separation weakened, with detection rates dropping from perfect classification to partial detectability. This pattern suggests that the broad tier structure remains stable across settings, but borderline cases are sensitive to whether Grok evaluates poems in a shared conversational context or as isolated items. For completeness,

the corresponding confusion matrices for both Grok evaluation settings are reported in the Appendix B.

Source Model	AI detected	AI detection rate	Human correct
ChatGPT-5	0/12	0.0%	12/12
Gemini 2.5 Pro	0/12	0.0%	12/12
StableLM Gamma-7B	0/12	0.0%	12/12
Gemma-2b-jpn	12/12	100.0%	12/12
LLaMA2	12/12	100.0%	12/12
LLM-JP-Instruct4	12/12	100.0%	12/12
Human-written baseline	—	—	12/12

Table 9.1. Grok’s performance in the *all haiku together* setting. “AI detected” indicates the number of AI-generated haiku correctly classified as AI out of the 12 haiku associated with each source model. “Human correct” indicates the number of human-written haiku correctly classified as human.

Source Model	AI detected	AI detection rate	Human correct
ChatGPT-5	0/12	0.0%	12/12
Gemini 2.5 Pro	0/12	0.0%	12/12
StableLM Gamma-7B	0/12	0.0%	12/12
Gemma-2b-jpn	8/12	66.7%	12/12
LLaMA2	7/12	58.3%	12/12
LLM-JP-Instruct4	5/12	41.7%	12/12
Human-written baseline	—	—	12/12

Table 9.2. Grok’s performance in the *one haiku at a time* setting. “AI detected” indicates the number of AI-generated haiku correctly classified as AI out of the 12 haiku associated with each source model. “Human correct” indicates the number of human-written haiku correctly classified as human.

9.1 Automated Classification Heuristics

A close reading of Grok’s justifications reveals that they rely on a relatively consistent set of classification heuristics. While these heuristics overlap with the cues reported by human respondents in open-ended questionnaire responses, they are applied with greater consistency and a more explicit explanatory vocabulary.

When Grok labels a haiku as AI-generated, it often highlights **seasonal inconsistency or weakness**. In these cases, the poem is said to lack a clear seasonal anchor, to misuse a *kigo*, or to combine imagery in a way that feels out of place in the seasonal setting. This emphasis on seasonality is consistent with the fact that haiku depends heavily on the effective use of seasonal references as part of its poetic logic. In Grok’s explanations, poems that fail to naturally ground themselves in a recognizable seasonal atmosphere are often considered artificial.

A second recurring heuristic concerns **mechanical or formulaic language**. Grok repeatedly describes AI-like items as sounding as though they have been put together on the surface only. Features associated with this include repetition, overly predictable word choice, tautological phrasing and keyword-like construction. Some haiku are characterised as merely “stating the obvious”, while others are said to rely on safe or generic seasonal imagery without producing a deeper poetic effect. This overlaps strongly with human open-ended responses, in which respondents often refer to repetition, excessive simplicity or the impression that words have been stitched together rather than composed intentionally.

A third major heuristic involves **coherence and interpretability**. Grok often interprets disjointed or unstable imagery as a sign of AI-generated content. AI-like haiku are described in their explanations as containing illogical combinations of images, abrupt transitions, incomplete thoughts or forced figures of speech. By contrast, human-like haiku are described as having more intentional imagery, a stronger internal connection between lines and more natural emotional or visual development. Again, this mirrors human responses, where difficulty of interpretation, broken flow and unclear intention were frequently cited as reasons for suspecting AI authorship.

A fourth heuristic concerns poetic depth versus surface plausibility. Grok distinguishes repeatedly between language that is merely acceptable on the surface and language that appears to be motivated by a deeper poetic purpose. Human-like haiku are often described as subtle, contemplative, evocative and emotionally restrained in a meaningful way. By contrast, AI-like haiku are more often described as technically proficient but lacking in poetic depth. This distinction is important because it suggests that Grok is checking not only for formal plausibility, but also whether the poem produces a layered poetic effect rather than a literal or decorative description.

At a general level, these heuristics bear a close resemblance to the cues reported by human participants in Chapter 8. There is a strong overlap in at least three areas. First, both Grok and the respondents rely on **rule-based reasoning**, particularly with regard to seasonality, structure, and poetic convention. Second, both associate **repetition and mechanical phrasing** with AI. Third, both treat **coherence, interpretability, and evocative imagery** as signals of humanness. The difference lies less in the type of cues than in the way they are applied. Human respondents apply them through varied and sometimes inconsistent personal interpretations, whereas Grok applies them in a more consistent and systematic manner from a rhetorical perspective.

However, this apparent stability should be interpreted with caution. Grok's explanations are generated by a language model and therefore reflect stylistic biases learned from the data, rather than providing a transparent symbolic analysis of haiku. Its justifications often sound highly confident, even when the classification outcome is incorrect. This is particularly pertinent in cases where high-quality AI-generated haiku are consistently misclassified as the work of humans. In these instances, rather than revealing hidden detectability, Grok's explanations demonstrate how easily a fluent and coherent AI-generated haiku can satisfy the same cues that Grok itself associates with human poetry.

9.2 Performance Across Model Tiers

The Grok evaluation reveals significant asymmetry in performance across model groups. Rather than displaying a gradual continuum of detectability, the results indicate a division between models that are more detectable and those that are much less so.

ChatGPT-5, Gemini 2.5 Pro, and StableLM Gamma-7B occupy the least detectable end of the spectrum. In the one-by-one evaluation, Grok classified all twelve AI-generated haiku from each of these models as 'Human'. This corresponds to a detection rate of 0% for all three systems. In other words, in the current evaluation setting, Grok failed to identify any of the haiku as having been written by AI. This is a striking result, as it suggests that these models' outputs satisfy the same criteria that Grok recognises as signs of human authorship: coherent imagery, natural seasonal integration and sufficient poetic plausibility.

At the other end are **Gemma-2b-jpn**, **LLaMA2**, and **LLM-JP-Instruct4**. These models remain partially detectable. In the one-by-one setting, Grok correctly identified only some of the AI-generated haiku as AI-written. These values indicate neither complete detectability nor indistinguishability. Rather, they suggest that some of these models' outputs still contain residual regularities that an automated judge can exploit, at least under the

current dataset and prompting conditions.

The evaluation also included the **human-written haiku baseline**. In the one-by-one setting, Grok classified all of these haiku as 'Human', yielding a perfect classification rate for the human set within the present sample. While this result should not be overinterpreted, it does suggest that Grok did not simply default to labelling every item as AI-generated. In fact, it showed a strong preference for human-written poems that matched its expectations of poetic subtlety and formal coherence.

These results are consistent with the broader argument that has already emerged from human evaluations. Overall, human participants performed only slightly above chance, and their judgements were highly item-dependent. Grok demonstrates a comparable sensitivity to perceived poetic quality, though with a different distribution across model groups. The key point is that detectability is not uniform. While some models produce outputs with residual artefacts or weaker poetic integration, others generate haiku that fall almost entirely within the perceptual space associated with human authorship.

This asymmetry is particularly revealing when comparing Gamma-7B and Gemma-2b-jpn. Although both are open-source, Japanese-capable systems, Gamma behaves like the highest-end models in the Grok evaluation, with a full collapse of detectability, whereas Gemma remains partially detectable. This suggests that parameter count alone is not the only relevant factor, but rather that model scale, training quality and generation behaviour interact in complex ways. At the same time, however, the difference is broadly consistent with the general observation that smaller or less optimised models are more likely to leave traces that can be detected by an automated judge.

Although the one-by-one approach revealed some differences at item level compared to the earlier multi-haiku conversational evaluation, the overall pattern remained consistent: the strongest models consistently evaded detection, while smaller or less optimised models were partially detectable.

The two Grok settings therefore allow for a layered interpretation of model behaviour. Rather than treating all AI-generated haiku as belonging to the same category, the data suggest that current systems occupy different positions along a continuum of detectability. While some remain statistically or stylistically distinguishable, others are effectively absorbed into the human-like zone under realistic evaluation conditions.

9.3 Detectability Collapse in High-End Models

The most significant outcome of the Grok comparison is the apparent detectability collapse of the strongest models. In the present study, this collapse is evident in Grok's systematic failure to identify any of the haiku generated by ChatGPT-5, Gemini 2.5 Pro or StableLM Gamma-7B as AI-written when evaluated individually. In all three cases, Grok classified every poem as having been written by a human.

This pattern is conceptually important because it goes beyond mere misclassification. It suggests that, once haiku are sufficiently fluent, coherent, seasonally plausible and aesthetically acceptable, the usual attribution cues cease to function reliably. In such cases, the problem is not just that the detector occasionally makes mistakes. Rather, the generated outputs match the detector's own model of human-like poetry so closely that, under the current evaluation setting, the distinction between human and machine becomes operationally unstable.

These results align with those of the human evaluation. Human respondents' judgements were already close to chance for ChatGPT-5, Gemini and Gamma, and Chapter 8 showed that these judgements were mainly guided by perceived poetic quality rather than diagnostic indicators of source. The Grok results extend this finding by demonstrating that even an automated evaluator with a consistent explanatory vocabulary makes the same mis-

take. When the poem satisfies the expected cues of human poetry, both human readers and an LLM-based judge recognise human authorship.

This collapse can be understood in the context of broader literature on the detection of machine-generated text. As language models become more capable of matching human distributions, the stylistic and distributional markers of artificiality weaken and become less stable, particularly in realistic conditions with limited sample sizes. In this case, the advanced models appear to produce haiku that no longer trigger Grok's AI-oriented heuristics: the seasonality seems coherent, the imagery appears deliberate, and the wording is not formulaic enough to be identified as machine-generated.

In contrast, detectability does not completely disappear for Gemma, LLaMA2 and Instruct4. Their outputs still contain enough regularities for Grok to classify some of them as AI-generated. This residual detectability may be due to weaker integration of imagery, less subtle seasonal grounding, flatter compositional development, or traces of mechanical structure. Nevertheless, even here, the distinction is far from clear-cut, suggesting that the collapse is not an all-or-nothing phenomenon, but rather the endpoint of a continuum.

A particularly important implication of this result is methodological. If both human and automated judges rely on broadly similar cues, such as coherence, seasonality, imagery and poetic depth, and if the strongest generators increasingly satisfy those cues, then authorship attribution becomes unstable, not because evaluators are careless, but because the task itself becomes less well-defined. In these circumstances, the distinction between "human-like" and "AI-like" writing is no longer based on clear, observable signals. Instead, it becomes a probabilistic judgment shaped by expectation and the residual artefacts that weaker systems still produce.

The Grok evaluation, conducted on a case-by-case basis, therefore strengthens the central claim of this thesis. High-end language models do more than generate acceptable haiku; they generate haiku that evade both human and machine attribution in a systematic way. Once this point is reached, detectability ceases to be a reliable indicator of authorship, instead reflecting the generator's remaining weaknesses. In this sense, the strongest models are not just harder to detect; they render the attribution problem unsolvable.

Nevertheless, a number of limitations should be noted. Firstly, even though the Grok comparison covers the full set of haiku used in the questionnaire study, it remains exploratory. Secondly, the number of items per model is still limited, meaning that strong separations between model groups may partly reflect features of the current dataset. Thirdly, Grok's justifications are themselves generated by a language model and should therefore be interpreted as a model-dependent explanatory style rather than as direct access to a transparent decision-making process. For these reasons, the present analysis is best interpreted as a comparative probe into detectability patterns rather than as a definitive benchmark for authorship detection.

Overall, the Grok comparison reveals that AI-generated haiku currently fall into at least two distinct categories. In one of these, represented by Gemma, LLaMA2 and Instruct4, traces of artificiality remain partially detectable. In the other regime, represented by ChatGPT-5, Gemini and Gamma, these traces become so faint that they cannot be attributed by either human or automated judges. This distinction provides a useful perspective on the transitional state of contemporary creative language generation. While AI outputs can still vary in detectability, the most advanced systems already reveal the limitations of using authorship recognition as an evaluation framework.

Chapter 10

Discussion and Implications

This chapter discusses the broader significance of the results presented in previous chapters, summarizing the main quantitative, qualitative, and comparative findings of the thesis. The overall aim of this research was to investigate whether contemporary AI systems can produce artefacts that are perceived as creative and, if so, to what extent this apparent creativity can approach or reproduce the effects typically associated with human artistic production. Japanese haiku was used as the empirical test case for this purpose because it offers an informative combination of strict formal constraints, cultural conventions, and interpretive openness.

Within this broader framework, the thesis addressed two more specific research questions. The first was **whether contemporary large language models could generate Japanese haiku that were both linguistically and poetically plausible** under a shared prompting framework. The second concerned **whether human readers could reliably distinguish these haiku from poems written by humans**. The results suggest that, while these two questions are related, they are not identical. Several models can now produce haiku that are perceived as coherent, readable and aesthetically plausible, yet this surface plausibility does not translate into stable or accurate authorship recognition.

Taken together, the descriptive analyses, inferential models, open-ended responses, and Grok comparison all point to the same conclusion: judgments about attribution in this field are guided more strongly by perceived poetic quality than by reliable indicators of the poem's true origin. However, the findings do not support the simple claim that AI-generated and human-written haiku are indistinguishable in every case. Rather, the results paint a more nuanced picture, showing that detectability varies between models, individual poems, and evaluation settings. The following sections discuss the implications of this pattern for computational creativity research and the use of authorship attribution as an evaluative framework.

10.1 Are AI Haiku Indistinguishable?

This thesis poses a key question: can AI-generated haiku be considered indistinguishable from those crafted by humans? The results suggest that the answer is neither straightforwardly yes nor no. Rather, indistinguishability appears to be a conditional and model-dependent phenomenon.

At the level of the human questionnaire, the study's strongest systems, ChatGPT-5, Gemini 2.5 Pro and StableLM Gamma-7B, produced recognition accuracy values close to chance. In practical terms, this means that respondents were generally unable to reliably distinguish their haiku from those written by humans. In contrast, Gemma, LLaMA2 and LLM-JP-Instruct4 were moderately detectable. This suggests that AI-generated haiku do not form

a single category. Some outputs still exhibit traces of artificiality that can be discerned by readers, while others no longer provide consistent indicators of machine authorship under normal evaluation conditions.

However, this distinction should be interpreted with caution. Item-level analyses revealed that recognition performance varied significantly across individual poems. While some haiku were correctly classified by the majority of participants, others were systematically misclassified. This suggests that detectability is not a consistent property of the entire model. Instead, it depends heavily on the specific poem being evaluated. Therefore, a more accurate question is not simply whether a model is distinguishable, but under what conditions, for which outputs and for which readers.

The Grok comparison reinforces this conclusion. In both evaluation settings, ChatGPT-5, Gemini and Gamma were less susceptible to automated detection than Gemma, LLaMA2 and Instruct4. At the same time, the strength of the separation depended on the context. When haiku were presented together, Grok produced a clearer distinction between the stronger and weaker model groups. However, when haiku were presented individually, the distinction weakened for the smaller or less optimised models. This suggests that indistinguishability is dependent not only on the model, but also on the evaluation.

For this reason, it is more accurate to say that, under realistic judgement conditions, some AI-generated haiku have become operationally indistinguishable, rather than absolutely indistinguishable in principle. The present findings suggest that high-end models can generate haiku that both human readers and an external, LLM-based judge fail to attribute reliably. This does not mean that no signal remains, but it does suggest that it has become weak enough to be practically indistinguishable.

10.2 Perceptual Heuristics vs Diagnostic Cues

A major conceptual contribution of the present study is the distinction between **perceptual heuristics** and **diagnostic cues**. The results show that respondents did not judge authorship at random. In fact, both open-ended responses and inferential analyses suggest that participants relied on consistent internal criteria to determine whether a haiku was written by a human or generated by AI.

These criteria included coherence, clarity of meaning, haiku-like wording, evocative imagery, emotional resonance, and the apparent appropriateness of the seasonal reference. In Chapter 8, respondents repeatedly referred to these features as the basis for their decisions. Poems judged as human-written were described as vivid, meaningful, emotionally effective, and natural in their phrasing. In contrast, poems judged as AI-generated were more likely to be described as repetitive, overly direct, awkwardly assembled or semantically unclear.

However, the inferential models showed that these cues were much more strongly related to *perceived authorship* than to *correct recognition*. This is a crucial distinction. A perceptual heuristic is a cue that readers find psychologically plausible and use consistently when making judgements. By contrast, a diagnostic cue is one that actually corresponds to the true source of the poem. In the present study, many of the cues used by respondents were meaningful and interpretable, but only weakly diagnostic.

This helps to explain why recognition accuracy remained low, despite the fact that respondents could often articulate clear reasons for their decisions. Their reasoning was not arbitrary; it was simply based on signals that had become unreliable indicators of authorship. Readers often judged a coherent and evocative poem to be human because they associated such qualities with human creativity. However, these qualities could also be produced by stronger language models. Conversely, a poem that felt flat, mechanical or forced was often deemed to be AI-generated, even when it had actually been written by a human.

The same pattern emerges in the Grok evaluation. Grok also relied on signals such as

seasonality, coherence, mechanical phrasing and poetic depth. This suggests that human readers and machine judges may converge on broadly similar evaluative heuristics, even if they are not consistent. However, convergence of heuristics does not guarantee alignment with the truth. In this domain, the attribution problem therefore arises not from an absence of judgment, but from a mismatch between the cues available to evaluators and those that genuinely distinguish the source.

10.3 The Halo Effect in Creative Attribution

Another key finding of this study is the presence of a halo-like pattern in creative attribution. In this context, this does not mean that participants first evaluated the aesthetic quality of a poem and then, on that basis, decided who had written it. Since the authorship judgment question came before the Likert-scale impression ratings in the questionnaire, such a causal interpretation would not be warranted. Rather, the results show that judgments of authorship and positive evaluations of the poem were closely associated with one another.

Inferential analyses revealed that impression variables such as favourability, haiku-like wording, coherence, fluency and related factors were strongly associated with poems being judged as human-written. However, these same variables did not reliably predict whether the judgment was correct. This suggests that judgments of authorship and evaluations of poetic quality were not independent in participants' responses. Poems judged as human-written also tended to receive higher impression ratings, indicating a halo-like association between perceived poetic success and human authorship.

The qualitative results support this interpretation. Respondents often justified their human classifications by referencing emotional subtlety, scene construction, poetic atmosphere and expressive depth. Conversely, when a poem felt flat, overly explicit or mechanically assembled, this was often taken as evidence of AI authorship. These patterns suggest that participants associated poetic value with humanness at the level of judgment, even if the questionnaire structure does not allow us to claim a direct one-way influence from aesthetic evaluation to source attribution.

This halo-like pattern is particularly important because it helps to explain why perceived humanness and correct recognition diverged so significantly. Once a model can produce fluent, coherent, evocative and seasonally plausible outputs, these outputs are more likely to be judged as human and also to receive more positive evaluations, even if they are machine-generated. In these circumstances, success in poetic imitation does not merely improve surface quality; it also destabilises the attribution process by making the cues associated with humanness easier for AI systems to satisfy.

This observation also links the current findings to wider discussions about computational creativity. Much of the existing literature considers whether AI-generated artefacts are genuinely creative or merely appear to be. The present study suggests that, at least in the context of haiku, perceived creativity and perceived authorship can become closely linked in human judgment. A poem judged as creative or poetically successful is also more likely to be judged as human-written. This makes source attribution particularly vulnerable to evaluative bias and suggests that perception of creativity cannot be cleanly separated from perception of authorship.

10.4 Implications for Authorship Evaluation

The results of this thesis have broader implications for the use of authorship evaluation in the study of AI creativity. In many discussions of generated text, the ability or inability of readers to distinguish between human and machine-generated outputs is considered a

measure of success [16], [7], [8]. However, the present findings suggest that this benchmark is becoming increasingly unstable.

Firstly, the study suggests that attribution of authorship is no longer a reliable standalone measure of creative distinction for high-performing systems. When both human respondents and an external, LLM-based judge fail to identify the source of a strong, AI-generated haiku, detectability begins to reflect the residual weaknesses of weaker models rather than being a robust property of machine-generated poetry in general. In this sense, the question “can readers tell?” becomes a moving and potentially misleading target.

Secondly, the findings suggest that evaluative frameworks should distinguish more clearly between **artifact quality** and **source attribution**. A poem may be aesthetically effective regardless of whether readers can identify its origin. Conversely, a poem may be attributed incorrectly, even when readers are highly confident in their judgment. Therefore, treating attribution as a measure of quality risks combining two different questions into one: whether the poem is good and whether it seems to be the work of a human author.

Thirdly, the study suggests that future evaluations of AI creativity should be more sensitive to the specific domain. Haiku is a highly structured poetic form in which seasonality, compression, ambiguity and suggestiveness all play a central role. In other creative fields, such as visual art, music or long-form literary writing, the cues used by observers may differ considerably. The broader implication is that creativity evaluation cannot be fully standardized across all domains. Instead, evaluations must take into account the specific conventions, expectations and interpretive habits associated with each form.

Fourthly, the Grok comparison suggests that automated judges do not solve the attribution problem straightforwardly. Although Grok applied its cues more consistently than human respondents, it ultimately relied on similar kinds of signals and produced similar results for the strongest models. This implies that replacing human readers with LLM-based judges does not solve the underlying issue. It merely reproduces it in another form.

Taken together, these points suggest that future evaluation frameworks may need to move beyond attribution-centered designs. More useful approaches could include distinguishing between process and product, comparing the reading of experts and non-experts, examining revision and intentionality, and evaluating how artefacts function within specific cultural and aesthetic traditions, rather than primarily asking whether the source can be guessed.

10.5 Limitations

Several limitations of the present study should be acknowledged.

Firstly, the number of haiku per model was limited. While the design was balanced and sufficient to reveal clear patterns, a larger set of items would have enabled more robust estimation of detectability and reduced the influence of poem-specific effects. This is particularly important because variability at the item level was one of the study’s clearest results.

Secondly, the experiment focused on a single creative domain and poetic form. While haiku are particularly well-suited to this type of analysis due to their combination of strict formal constraints and high interpretive openness, this also limits generalization. Therefore, the present findings cannot automatically be extended to other genres of poetry or prose, or to other creative domains such as music or visual art.

Thirdly, the sample of participants consisted mainly of university students in Japan. This was a deliberate and methodologically appropriate choice, given that the study focused on Japanese haiku written in the original language, which required respondents with direct exposure to Japanese linguistic and cultural conventions. However, this also limits the generalisability of the findings to other populations. Notably, familiarity with haiku varied sub-

stantially among respondents, and qualitative evidence suggests that readers with different levels of literary knowledge sometimes interpreted the same textual properties differently.

Fourthly, the closed-source models were evaluated through interface-based generation rather than a fully standardized, programmable pipeline. While this reflects practical constraints and does not invalidate the comparison, it does mean that the generation conditions for these systems were not as tightly controlled as those for the open-source models.

Fifthly, the Grok comparison remains exploratory. Although it was applied to the entire set of haiku used in the questionnaire and revealed an informative tier structure, the results are still based on a relatively small sample and one specific external model acting as judge. Therefore, its results should be interpreted as evidence of comparative detectability patterns rather than as a definitive benchmark for automated attribution.

Finally, a broader conceptual limitation also needs to be considered. As generative systems improve, the task of attributing authorship may become less stable. If the strongest models can produce texts that closely match the cues that readers usually associate with human creativity, low recognition accuracy may not only be due to the limitations of the present study. It may also reflect the fact that the differences between human- and AI-generated texts are becoming less observable. In this sense, the limitation may lie not only in the study design, but also in the attribution framework itself.

Overall, the present findings suggest that contemporary haiku generation has reached a stage at which attribution accuracy alone is an inadequate means of evaluating the strongest systems. The next chapter concludes the thesis by summarizing the main contributions, reflecting on the broader theoretical significance of the results, and considering future research directions in AI creativity and human-machine comparisons.

Chapter 11

Conclusion

This thesis investigated the creative potential of contemporary Large Language Models by generating and evaluating Japanese haiku. The study examined both the poetic quality of AI-generated haiku and the extent to which readers can distinguish them from human-written poems by combining a controlled generation pipeline, a human questionnaire study, inferential statistical modeling, qualitative analysis of open-ended responses, and a comparative evaluation using Grok as an external judge.

The broader aim was to test whether current models could imitate a poetic form that was both culturally and structurally constrained, and to understand what happens when readers are asked to infer authorship from short creative texts. In this sense, the thesis addressed both generation and evaluation problems. The findings suggest that these two issues are deeply intertwined: as models become more capable of producing fluent, coherent and aesthetically plausible poems, the cues that readers use to determine authorship become more unstable.

11.1 Main Findings

Several main findings emerge from the study.

Firstly, the results demonstrate that contemporary language models vary significantly in their capacity to produce Japanese haiku that resemble human creations. The strongest models, ChatGPT-5, Gemini 2.5 Pro and StableLM Gamma-7B, produced haiku that human respondents generally found difficult to distinguish from human-written ones. However, their recognition accuracy remained close to chance, suggesting that under realistic judgement conditions, these systems can generate poems that occupy much of the same perceptual space as human-authored haiku. In contrast, Gemma, LLaMA2 and LLM-JP-Instruct4 were more easily detectable overall, though not completely so.

Secondly, it was found that detectability was strongly item-dependent. Recognition performance was inconsistent across all outputs from the same model. While some poems were consistently classified correctly, others were systematically misclassified. This suggests that the attribution of authorship depends not only on the model that generated the poem, but also on the specific local features of each haiku in isolation. Consequently, detectability cannot be considered a fixed, global property of an entire system.

Thirdly, the inferential analyses revealed a clear distinction between perceived human-ness and correct recognition. Impression ratings, particularly those relating to favourability, haiku-like wording, coherence and related variables, were strongly associated with whether a poem was judged to have been written by a human. However, these same cues were much less effective in predicting whether the judgement was correct. This suggests that readers were not deciding randomly. They were applying meaningful internal criteria, but these did

not reliably correspond to the true source of the text.

Fourthly, the qualitative analysis clarified what those criteria entailed in practice. Participants frequently mentioned coherence, clarity, imagery, emotional resonance, the naturalness of phrasing and rule-based intuitions about haiku structure and seasonal words. However, the open-ended responses showed considerable variability in how these cues were interpreted. More knowledgeable readers sometimes viewed ambiguity as poetic depth, whereas less experienced readers tended to view the same ambiguity as confusion or artificiality. This suggests that judgements about authorship are filtered through culturally and personally shaped expectations about what a poem written by a human should sound like.

Fifthly, the Grok comparison reinforced these findings by demonstrating that an external language model acting as judge relied on similar types of cues. Grok was more successful at distinguishing weaker or less optimised models than the strongest ones, but systematically failed to detect AI-generated haiku from ChatGPT-5, Gemini and Gamma. This suggests that the attribution problem is not confined to human readers. Even an automated evaluator with a more consistent vocabulary of explanations can make the same mistake when strong models satisfy the cues associated with human poetry.

Taken together, these results suggest that the most effective models do more than just produce formally acceptable haiku. Rather, they produce haiku that can be interpreted as sufficiently human in style, atmosphere, and poetic plausibility by both readers and automated judges. In this sense, the attribution task weakens not because evaluators stop applying criteria, but because the available criteria cease to be reliably diagnostic.

11.2 Theoretical Implications

The findings of this thesis have several theoretical implications for the study of computational creativity.

The first implication relates to the relationship between creativity and attribution. The results suggest that, in human judgement, perceived creativity and perceived humanity are closely linked. Poems that appear coherent, evocative, emotionally resonant or aesthetically successful are more likely to be attributed to a human author. This indicates that evaluations of source and evaluations of quality are not independent of each other. Instead, they interact through a halo-like mechanism, whereby positive aesthetic judgements spill over into assumptions about origin. This has important consequences for computational creativity research, because it shows that the question "who wrote this?" is partly shaped by the question "how good does this feel?"

A second implication relates to the distinction between creativity itself and how creativity is perceived. As discussed in the theoretical background, Colton argues that the two are not the same. The present study strongly supports this view. A poem may satisfy many of the perceptual conditions that cause readers to interpret it as creative or human-like, even if these perceptions do not reveal anything stable about its source. In other words, the evaluation of creativity cannot be reduced to surface plausibility alone. The strongest models in this thesis produced poems that were often perceived as having been written by a human author because they satisfied readers' expectations about poetic language, rather than because readers had access to genuinely reliable signals of authorship.

A third implication relates to the limitations of formalising creativity. The results suggest that certain aspects of creative production can be understood by examining observable features such as coherence, structure, novelty and aesthetic plausibility. However, the study also shows that these dimensions do not fully capture what readers perceive as creativity, particularly in domains such as haiku, where ambiguity, subtlety, emotional restraint, and cultural resonance are key. This indicates that creativity can only be operationalised partially, and always within a specific evaluative framework.

A fourth implication is that creativity may need to be studied separately in different creative domains. This thesis focuses on haiku, a form in which compression, seasonality, suggestion and interpretive openness are all key features. In other creative domains, such as music, painting or narrative prose, the cues that shape perceived creativity may be quite different. Therefore, the evaluation of AI creativity should not aim for one universal standard that is detached from the conventions of the domain in question. Rather, it should acknowledge that what constitutes creativity depends partly on the structure of the artefact, the tradition to which it belongs, and the interpretive expectations of its audience.

A fifth implication relates to the evolving cultural perception of AI itself. The open-ended responses suggest that some respondents may still associate AI with repetitive, shallow or obviously flawed language, despite the fact that contemporary systems are often much more advanced. If so, part of the judgement of authorship may be shaped by an outdated mental model of what AI-generated text is supposed to look like. This suggests that judgements about attribution are historically situated and influenced by shifting social expectations about technology. As AI systems continue to improve, human evaluators may increasingly rely on assumptions that no longer reflect the actual properties of the generated text.

Overall, the theoretical significance of this thesis lies in demonstrating that the evaluation of AI creativity cannot be reduced to simple dichotomies such as human versus machine, authentic versus artificial, or creative versus imitative. The findings instead suggest a more complex picture, in which creativity, plausibility, authorship and evaluation are intertwined through perception, expectation and cultural framing.

11.3 Future Research Directions

The present study opens several directions for future research.

The first area of focus is the conceptual problem of creativity itself. While the present thesis used haiku as a concrete domain in which to test the creative potential of current language models, it also raises a broader question for future research: to what extent can creativity be defined in a precise enough way to be modelled, taught or evaluated computationally? The results suggest that while some components of creativity can be approximated through dimensions such as novelty, coherence, formal structure and aesthetic value, these do not exhaust the phenomenon. In poetry, for instance, creativity may also encompass suggestion, ambiguity, emotional restraint, and cultural resonance. Therefore, future work should examine whether computational creativity is best understood through a set of domain-specific operationalisations rather than one universal definition.

A second area of focus is the evaluation process itself. The present findings suggest that attribution of authorship is becoming an increasingly unreliable benchmark for high-performing systems. Future work should therefore explore alternative evaluation designs that more clearly separate artefact quality from source identification. For instance, studies could compare blind quality judgements with judgements informed by the source, measure how evaluations change when respondents are informed that a poem was written by AI or a human, or examine how attribution differs between expert and non-expert readers. These designs would clarify whether AI-generated artefacts are evaluated on their own merits or based on preconceived notions about their creators.

A third area of focus is the role of the reader's background. The qualitative results of this thesis suggest that respondents with varying degrees of literary familiarity interpret the same features differently. Future research could examine this systematically by comparing groups with different types of expertise, such as haiku poets, literature students, general readers and readers with a strong technical knowledge of AI systems. This would help to determine whether greater familiarity with poetry or AI, or both, changes the way in which

creative artefacts are judged.

A fourth area of focus is the cultural and linguistic context. As haiku is deeply rooted in Japanese poetic traditions, future research could examine how the same poems are evaluated by Japanese and non-Japanese readers, or how AI-generated haiku perform in translation. Similar cross-cultural studies could investigate whether human-like evaluations depend on specific cultural conventions that strong models learn to imitate or whether they generalise more broadly across audiences.

A fifth area of focus is automated judges. The Grok comparison revealed that LLM-based judges may employ heuristics similar to those of humans, resulting in comparable failures when confronted with sufficiently plausible outputs from stronger models. Therefore, future research should examine more systematically whether automated judges can meaningfully contribute to creativity evaluation and under what conditions. This should include comparing multiple LLM-as-judge systems, testing their stability across prompts and contexts, and investigating whether they provide genuine evaluative insight or simply reproduce learned rhetorical expectations.

Ultimately, future research should consider creativity as a property of both outputs and processes. A poem may appear creative without revealing the process by which it was generated. Therefore, one promising direction would be to move beyond evaluating only the final product and examine revision behaviour, interaction, intentionality, and the role of prompting and iterative selection in the creative process. Such an approach may help to distinguish between mere stylistic plausibility and richer forms of generative agency.

Although these directions originate from a study of Japanese haiku, many of them extend beyond poetry. Questions concerning the operationalisation of creativity, audience expectations, authorship attribution and the evaluation of AI-generated artefacts are likely to extend to other creative domains, such as visual art, music and narrative writing.

In conclusion, this thesis has demonstrated that contemporary large language models can generate haiku that are frequently considered to be human-like. This challenges the effectiveness of authorship attribution as a straightforward evaluative tool. However, the findings do not imply that human and machine creativity are identical, nor that the concept of creativity has become obsolete. Rather, they suggest that the boundaries through which creativity is recognised are shifting. Understanding these changes and developing better ways to evaluate them will be an important task for future research in computational creativity and human–AI comparisons.

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Appendix A

Prompt and Decoding Experiment Logs

Settings for one-shot beam search haiku generation:

Script	Prompt Style	few_shot_k	num_targets	num_samples	Sampling	Beam Search	Diversity Penalty	Nucleus Sampling
Prova 1	Basic template	4	10	100	Temperature=0.7, do_sample=True	No	None	None
Prova 2	Basic template	4/8	10/30	100/300	Temperature=0.7, do_sample=True	Yes, num_beams=5	None	None
Prova 3	Optimized prompt (ChatGPT help)	4	10	100	No (beam search only)	Yes, 100 beams, 10 groups	1.0	None
Prova 4	Optimized prompt + semantic retrieval	8	30	300	Top-p=0.9, Temperature=0.7	Yes, num_beams=5	None	semantic k-NN
Prova 5	Simplified	4	30	50	Beam search only	50 beams, 5 groups	0.7	None
Prova 6	Optimized + BLEU	8	30	5	Beam search only	5 beams, no groups	None	None
Prova 7	Optimized + 5-7-5 + BLEU rerank	5	20	6	Beam search only	6 beams, 2 groups	0.5	None
Prova 8	Gemma-style prompt	5	20	6	Beam search only	6 beams, 2 groups	0.5	None
Prova 9	Gemma-style + GPT-3.5 scoring (NOT TRIED)	5	20	6	Beam search only	6 beams, 2 groups	0.5	None
Prova 10	Gemma-style + author diversity	5	20	6	Beam search only	6 beams, 2 groups	0.5	None
Prova 11	Gemma-style + constraints + removed spaces	6	20	6	Beam search only	6 beams, 3 groups	0.7	None
Prova 12	Gemma-style + constraints + removed spaces	4	40	4	Beam search only	4 beams, 2 groups	0.4	None
Prova 13	Same as 11 on ELYZA (LLaMA)	6	20	6	Beam search only	6 beams, 3 groups	0.7	None
Prova 14	Same as 11 on STABLELM	6	20	6	Beam search only	6 beams, 3 groups	0.7	None
Prova 15	Gemma-style prompt optimized by Claude v1	6	20	6	Beam search only	6 beams, 3 groups	0.7	None
Prova 16	Gemma-style prompt optimized by Claude v2	6	20	6	Beam search only	6 beams, 3 groups	0.7	None
Prova 17	Gemma-style prompt optimized by Claude v3	6	20	6	Beam search only	6 beams, 3 groups	0.7	None
Prova 18	PromptOptimizer loop (automatic template scoring)	6	20	5	Beam search only	6 beams, 3 groups	0.7	None

Script	Retrieval Method	Special Token Handling		BLEU	max_new_tokens
Prova 1	Same season & kigo	No	No		20
Prova 2	Same season & kigo	Yes, token added	No		50
Prova 3	Same season & kigo	Yes, token with stopping	No		50
Prova 4	Semantic k-NN on embeddings	Yes, token with stopping	No		50
Prova 5	Same season & kigo (excluding target)	Yes, with stopping	No		30
Prova 6	Same season & kigo (excluding target)	Yes, with stopping	Yes		20
Prova 7	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		40
Prova 8	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		40
Prova 9	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		40
Prova 10	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		40
Prova 11	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		32
Prova 12	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		20
Prova 13	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		20
Prova 14	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		20
Prova 15	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		20
Prova 16	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		20
Prova 17	Same season & kigo (excluding target)	Yes, with stopping	Yes, used for rerank		20
Prova 18	PromptOptimizer (best of 4 templates)	Yes, with stopping	Yes (used in prompt scoring & rerank)		20

Settings for beam search and both one-shot and line-by-line haiku generation:

Script (ID)	Prompt style	few_shot_k	num_targets	num_samples	Sampling	Beam search	Diversity penalty	Nucleus sampling
iterative_eval_1 (T1)	Gemma-style examples + author/season/kigo header (no hint)	6	20	5 (L2) + 5 (L3 per L2)	Beam only	6 beams, 3 groups (for L2/L3)	0.7	None
iterative_eval_mora (T2)	Gemma-style + per-step labels; more-constrained; debug cleaners	6	20	20 per step	Beam only	20 beams, 0 groups	-	None
iterative_eval_mora_2 (T3)	As T2; improved extraction; mora-constrained	6	20	20 per step	Beam only	20 beams, 0 groups	-	None
haiku_beamsearch_STABLELM_Gemma (revised) (T4)	Gemma-style + explicit "5-7-5" scaffold; optional hint; \<example_end>	6	20	5 (L2) + 5 (L3 per L2)	Beam only	6 beams, 3 groups (for L2/L3)	0.7	None
fewshot_eval_mixed_pool (T5)	Gemma-style; optional hint	6	20	5	Beam only	6 beams, 3 groups	0.7	None
fewshot_eval_mixed_filtered (T6)	Gemma-style; optional hint	6	20	5	Beam only	6 beams, 3 groups	0.7	None
fewshot_eval_mixed_filtered_2 (T7)	Gemma-style; optional hint	6	20	5	Beam only	6 beams, 3 groups	0.7	None
iterative_eval_2 (T8)	Gemma-style examples; header; prefer 5-7-5 in retrieval	6	20	5 (L2) + 5 (L3 per L2)	Beam only	6 beams, 3 groups (for L2/L3)	0.7	None

Script (ID)	Mora enforcement	Retrieval method	Special token handling	
T1	Report only (no enforcement)	Same season & kigo (excluding target); prefer same author → fallback season+kigo → season	\<end_of_turn> added + stopping	
T2	Per-line enforced (5/7/5)	Same as T1	\<end_of_turn> added + stopping	
T3	Per-line enforced (5/7/5)	Same as T1	\<end_of_turn> added + stopping	
T4	Report only (no enforcement)	Same as T1; pool df_masters (= df copy)	\<example_end> for examples + \<end_of_turn> for stopping	
T5	Report only (no enforcement)	Mixed: 2x same-kigo (1/author) + 4x same-season (1/author), excluding target	\<end_of_turn> added + stopping	
T6	Report only (no enforcement)	Mixed + 5-7-5 preference: up to 3 strict same-kigo (1/author) + loose same-season (1/author), prefer 1s_575, staged fallback	\<end_of_turn> added + stopping	
T7	Report only (no enforcement)	Mixed (strict_k=2) + prefer 1s_575, with graceful fallback; 1/author	\<end_of_turn> added + stopping	
T8	Report only (no enforcement)	Season-first; prefer 5-7-5; 1/author then fill	\<end_of_turn> added + stopping	
Script (ID)	Reranking objective	Logged metrics	max_new_tokens	Example delimiter
T1	BLEU - 0.2-log(PPL) (full haiku)	BLEU, PPL, mora_rate, kigo_rate	~20/20/20 (L1/L2/L3)	\<end_of_turn>
T2	None (per-line greedy)	BLEU, PPL, mora_rate, kigo_rate	16/16/16	\<end_of_turn>
T3	None (per-line greedy)	BLEU, PPL, mora_rate, kigo_rate	16/16/16	\<end_of_turn>
T4	BLEU - 0.2-log(PPL) (full haiku)	BLEU, PPL, mora_rate, kigo_rate	10/14/10	\<example_end>
T5	BLEU - 0.2-log(PPL) (across 5 one-shot cands)	BLEU, avg PPL, mora_rate, kigo_rate	20 (one-shot)	\<end_of_turn>
T6	BLEU - 0.2-log(PPL) (one-shot)	BLEU, avg PPL, mora_rate, kigo_rate	20 (one-shot)	\<end_of_turn>
T7	BLEU - 0.2-log(PPL) (one-shot)	BLEU, avg PPL, mora_rate, kigo_rate	20 (one-shot)	\<end_of_turn>
T8	BLEU - 0.2-log(PPL) (full haiku)	BLEU, PPL, mora_rate, kigo_rate	20/20/20	\<end_of_turn>

Script ID	Generation mode	Decoding	Mora enforcement	L1 candidates	L2 per L1	L3 per L2	Reranking objective
T1	Line-by-line (3-step)	Beam search	Report only	1 (best single)	5	5	BLEU - 0.2*log(PPL) (full haiku)
T2	Line-by-line (3-step)	Beam search	Per-line enforced (5/7/5) 20 → pick first valid 5 (else first)	20 → pick first valid 5	20 → pick first valid 7	20 → pick first valid 5	None (per-line greedy)
T3	Line-by-line (3-step)	Beam search	Per-line enforced (5/7/5) 20 → pick first valid 5	20 → pick first valid 5	20 → pick first valid 7	20 → pick first valid 5	None (per-line greedy)
T4	Line-by-line (3-step)	Beam search	Report only	1 (best single)	5	5	BLEU - 0.2*log(PPL) (full haiku)
T5	One-shot (whole haiku)	Beam search	Report only	N/A	N/A	N/A	BLEU - 0.2*log(PPL) (across 5 one-shot cands)
T6	One-shot (whole haiku)	Beam search	Report only	N/A	N/A	N/A	BLEU - 0.2*log(PPL) (across 5 one-shot cands)
T7	One-shot (whole haiku)	Beam search	Report only	N/A	N/A	N/A	BLEU - 0.2*log(PPL) (across 5 one-shot cands)
T8	Line-by-line (3-step)	Beam search	Report only	1 (best single)	5	5	BLEU - 0.2*log(PPL) (full haiku)

Settings for Sampling with line-by-line haiku generation:

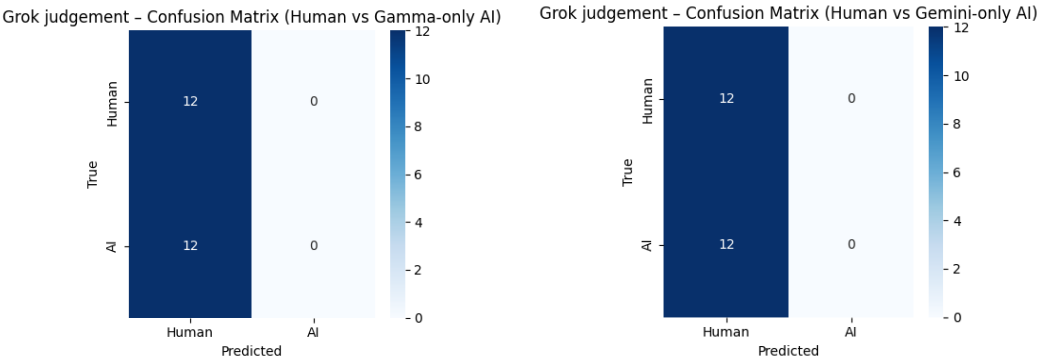
Script (ID)	Reranking objective			Logged metrics		max_new_tokens	Example delimiter
T9	-log(PPL) + 0.8-meter + 0.4-kigo - 1.2-max(BLEU_ex, BLEU_loc)			BLEU, PPL, mora_rate, kigo_rate		24/24/24 (per step)	\<end_of_turn>
T10	Same as T9 (on normalized text; simple quality gates)			BLEU, PPL, mora_rate, kigo_rate		24/24/24	\<end_of_turn>
T11	Same as T9 (continuation-only; forbidden set)			BLEU, PPL, mora_rate, kigo_rate		24/24/24	\<end_of_turn>
T12	Same as T11			BLEU, PPL, mora_rate, kigo_rate		24/24/24	\<end_of_turn>
T13	Same as T11 (kigo match may be phonetic)			BLEU, PPL, mora_rate, kigo_rate		24/24/24	\<end_of_turn>
T14	Same as T13, but similarity gate BLEU ignores kigo; L3 extractor prefers kigo; optional kigo_hint			BLEU, PPL, mora_rate, kigo_rate		24/24/24	\<end_of_turn>
T15	Same as T14 (BLEU ignores kigo; +kireji/kanji bonuses; -katakana/repeat penalties)			BLEU, PPL, mora_rate, kigo_rate		22/22/22	\<end_of_turn>
T16	Same as T15			BLEU, PPL, mora_rate, kigo_rate		22/22/22	\<end_of_turn>

Script ID	Generation mode	Decoding	Mora enforcement	L1 candidates	L2 per L1	L3 per L2	Reranking
T9	Line-by-line (3-step)	Sampling (T=0.7)	Per-line 5/7/5 (count_mora)	≤48 → first valid 5	≤48 → first valid 7	≤48 → 575+kigo+BLEU ok	Style (-log PPL - sim); accept first acceptable
T10	Line-by-line (3-step)	Sampling	Per-line 5/7/5 + bad-line filters	≤100 tries	≤100 tries	until acceptable	Style rerank on L3; reject dups/bad endings
T11	Line-by-line (3-step)	Sampling	Strict 5/7/5 per line	≤ max_line_attempts =100	idem	≤ max_haiku_attempts =200	Accept only if 575+kigo+BLEU; fallback = best_seen
T12	Line-by-line (3-step)	Sampling	Strict 5/7/5 + Unicode cleaning	idem	idem	idem	V3 + forbidden-set against prompt bleed
T13	Line-by-line (3-step)	Sampling	Strict 5/7/5	idem (100)	idem	cap 200	Adds phonetic kigo check; BLEU gate; style rerank
T14	Line-by-line (3-step)	Sampling	Strict 5/7/5	idem (100)	idem	cap 800	Prefer kigo in L3; BLEU-kigo; meter-aware fallbacks; log
T15	Line-by-line (3-step)	Sampling	Strict 5/7/5	idem (100)	idem	cap 800	Style: -log PPL + meter + kigo + kireji/kanji - sim; ordered fallbacks
T16	Line-by-line (3-step)	Sampling	Strict 5/7/5	idem (100)	idem	cap 800	V7 + stronger banned/regex guards ("行目" etc.) → slower

Appendix B

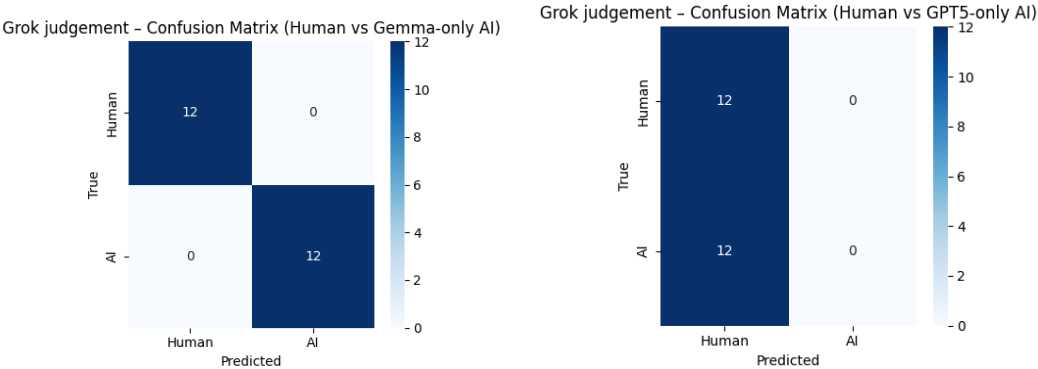
Grok Confusion Matrices

Grok confusion matrices: all haiku evaluated together



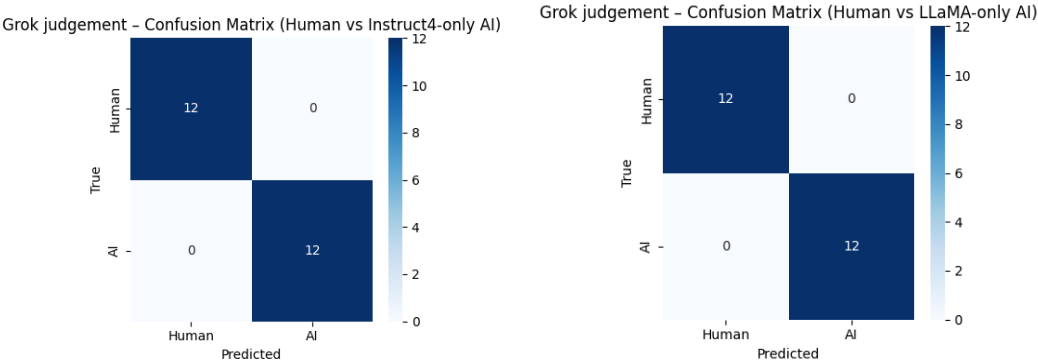
Confusion matrix for Grok’s evaluation of
Gemma-generated haiku

Confusion matrix for Grok’s evaluation of
Gemini-generated haiku



Confusion matrix for Grok’s evaluation of
Gemma-generated haiku

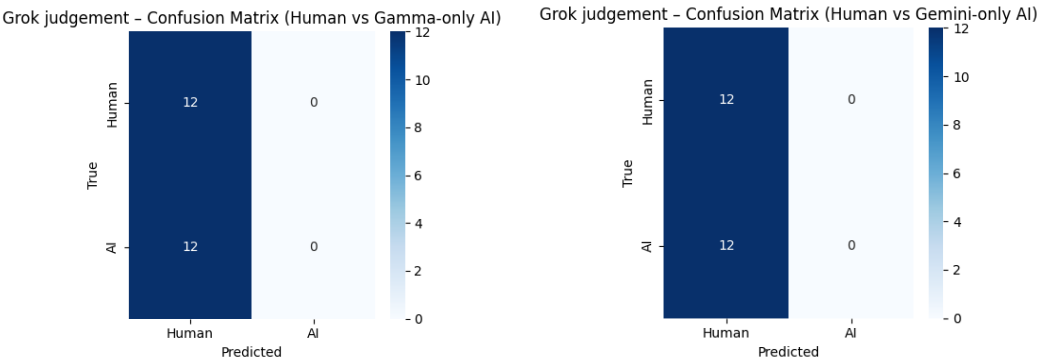
Confusion matrix for Grok’s evaluation of
ChatGPT-5-generated haiku



Confusion matrix for Grok’s evaluation of instruct4-generated haiku

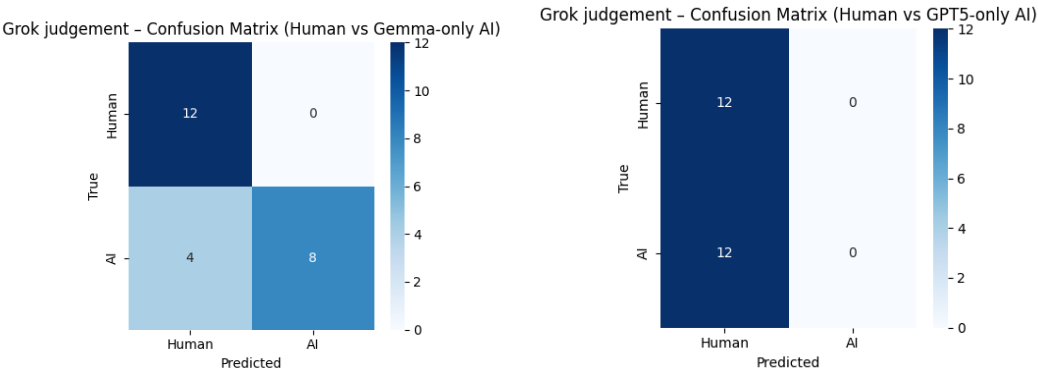
Confusion matrix for Grok’s evaluation of LLaMA2-generated haiku

Grok confusion matrices: haiku evaluated one at a time



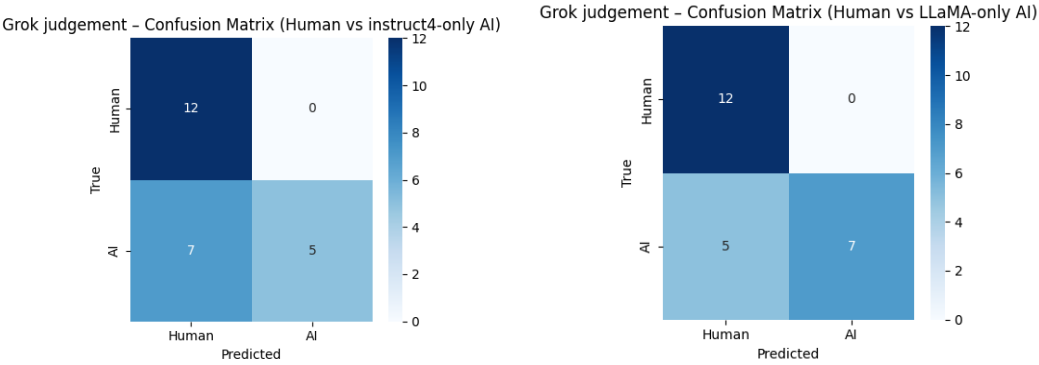
Confusion matrix for Grok’s evaluation of Gamma-generated haiku

Confusion matrix for Grok’s evaluation of Gemini-generated haiku



Confusion matrix for Grok’s evaluation of Gemma-generated haiku

Confusion matrix for Grok’s evaluation of ChatGPT-5-generated haiku



Confusion matrix for Grok’s evaluation of
instruct4-generated haiku

Confusion matrix for Grok’s evaluation of
LLaMA2-generated haiku